$$\omega_\epsilon := \begin{cases} \omega \setminus \bar{B}_\epsilon(\eta_0) & \text{for } \eta_0 \in \omega, \\ \omega \cup \bar{B}_\epsilon(\eta_0) & \text{for } \eta \in \Omega \setminus \overline{\omega}. \end{cases}$$

$$-\partial_t \bar{p}_\epsilon - \Delta \bar{p}_\epsilon = -2y^{\bar{u}_\epsilon, f, \epsilon} \quad \text{in } \Omega \times (0, T]$$

$$\min_{u \in L_2(0,T;L_2(\Omega))} J(\epsilon, u, f) = J(\epsilon, \bar{u}_\epsilon, f).$$

$$\lim_{\epsilon \searrow 0} \frac{J(0, \hat{\chi}_{\omega_\epsilon} \bar{p}_\epsilon, f) - J(0, \hat{\chi}_\omega \bar{p}_\epsilon, f)}{|B_\epsilon(0)|} = \frac{1}{4\gamma} \int_0^T (\bar{p}^{f,\omega}(\eta_0, t))^2 \, dt.$$

$$J(0, \hat{\chi}_{\omega_\epsilon} \bar{p}_\epsilon, f) = \int_{\Omega_T} y_\epsilon^2 + \frac{1}{4\gamma} \chi_{\omega_\epsilon} \bar{p}_\epsilon^2 \, dx \, dt,$$

$$\partial y_\epsilon - \Delta y_\epsilon = \frac{1}{2\gamma} \chi_{\omega_\epsilon} \bar{p}_\epsilon \quad \text{in } \Omega \times (0, T]$$

$$\int_0^1 \partial_y \mathcal{L}(\epsilon, s y_\epsilon + (1-s) y_0, \bar{q}_\epsilon; \bar{p}_\epsilon)(\psi) = 0 \text{ for all } \psi.$$

$$-\partial_t \bar{q}_\epsilon - \Delta \bar{q}_\epsilon = -(y_\epsilon + y_0) \quad \text{in } \Omega \times (0, T]$$

$$\lim_{\epsilon \searrow 0} \frac{J(0, \hat{\chi}_{\omega_\epsilon} \bar{p}_\epsilon, f) - J(0, \hat{\chi}_\omega \bar{p}_\epsilon, f)}{|B_\epsilon(0)|} = \int_0^T \frac{1}{4\gamma} (\bar{p}(\eta_0, t))^2 \, dt.$$

$$J(0, u, f) = \tilde{G}(0, u, f, y^{u,f,\omega}, p^{u,f,\omega}).$$

$$J(\epsilon, \bar{u}_\epsilon, f) - J(0, \bar{u}_\epsilon, f) = -\mathrm{sgn}_\omega(\eta_0) \int_0^T \int_{B_\epsilon(\eta_0)} \bar{u}_\epsilon \bar{q}_\epsilon \, dx \, dt.$$

$$J(\epsilon, \bar{u}_\epsilon, f) - J(0, \bar{u}, f) = J(0, \bar{u}_\epsilon, f) - J(0, \bar{u}, f) - \mathrm{sgn}_\omega(\eta_0) \frac{1}{2\gamma} \int_0^T \int_{B_\epsilon(\eta_0)} \chi_{\omega_\epsilon} \bar{p}_\epsilon \bar{q}_\epsilon \, dx \, dt.$$

$$\liminf_{\epsilon \searrow 0} \frac{J(\epsilon, \bar{u}_\epsilon, f) - J(0, \bar{u}, f)}{|B_\epsilon(\eta_0)|} \geq \frac{1}{4\gamma} \int_0^T (p^{\bar{u}, f, \omega}(\eta_0, t))^2 \, dt.$$

$$\mathcal{G}(\epsilon, f) := \mathcal{J}(\omega_\epsilon, f), \quad \ell(\epsilon) := |B_\epsilon(\eta_0)|.$$

$$\lim_{t \searrow 0} \frac{\mathcal{G}(t, f) - \mathcal{G}(0, f)}{\ell(t)} =: \partial_\ell \mathcal{G}(0^+, \bar{f})$$

$$\lim_{k \to \infty} \frac{\mathcal{G}(t_{n_k}, \bar{f}_{t_{n_k}}) - \mathcal{G}(0, \bar{f}_{t_{n_k}})}{\ell(t_{n_k})} = \partial_\ell \mathcal{G}(0^+, \bar{f}).$$

$$\frac{d}{d\ell} g_1|_{t=0} = \lim_{\epsilon \searrow 0} \frac{g_1(\epsilon) - g_1(0)}{|B_\epsilon(0)|} = \max_{f \in \mathfrak{V}_1(0)} \mathcal{T} \mathcal{J}_1(\omega, f).$$

$$G(\epsilon, \bar{u}^{f,\omega_\epsilon}, f) - G(0, \bar{u}^{f,\omega_\epsilon}, f) = -\mathrm{sgn}_\omega(\eta_0) \int_0^T \int_{B_\epsilon(\eta_0)} \bar{u}^{f,\omega_\epsilon} \bar{q}^{\bar{u},f,\omega_\epsilon} \, dx \, dt,$$

$$\bar{u}^{f,\omega_\epsilon}(t) := \bar{u}_\epsilon(t) = \frac{1}{2|\Omega|\gamma}\int_{\omega_\epsilon} \bar{p}^{\bar{u}_\epsilon,f,\omega_\epsilon}(t,x)\,dx$$

$$\lim_{\epsilon\searrow 0}\frac{G(\epsilon,\bar{u}^{\omega,f},f)-G(0,\bar{u}^{\omega,f},f)}{|B_\epsilon(\eta_0)|} = -\mathrm{sgn}_\omega(\eta_0)\int_0^T \bar{u}^{\omega,f}(t)\bar{p}^{\bar{u},f,\omega}(t,\eta_0)\,dt.$$

$$\partial_\ell G(0^+,\bar{u}^{\omega,f},f) = -\mathrm{sgn}_\omega(\eta_0)\int_0^T \bar{u}^{f,\omega}(t)\bar{p}^{\bar{u},f,\omega}(t,\eta_0)\,dt.$$

$$\lim_{k\to\infty}\frac{G(\epsilon_{n_k},\bar{u}_{\epsilon_{n_k}},f_{n_k})-G(0,\bar{u}_{\epsilon_{n_k}},f_{n_k})}{|B_{\epsilon_{n_k}}(\eta_0)|} = -\mathrm{sgn}_\omega(\eta_0)\int_0^T \bar{u}^{\omega,f}(t)\bar{p}^{\bar{u},f,\omega}(t,\eta_0)\,dt.$$

$$f_h = \sum_{j=1}^N f_j\phi_j\,,\quad f_j\in\mathbf{R}\,,\phi_j\in V_h\,,$$

$$\begin{aligned}\partial_t y-\Delta y &= \chi_\omega u && \text{in } \Omega\times(0,T],\\ y &= 0 && \text{on } \partial\Omega\times(0,T],\\ y(0) &= f && \text{on } \Omega.\end{aligned}$$

$$\varphi\circ T_t \to\varphi \quad \text{ strongly in } L_2(\Omega).$$

$$\frac{\varphi_n\circ T_{t_n}-\varphi_n}{t_n} \rightharpoonup \nabla\varphi\cdot X \quad \text{ weakly in } L_2(\Omega).$$

$$\mathcal{J}_1(\omega,f) = \inf_{u\in\mathrm{U}} J(\omega,u,f) = \int_0^T \|y(t)\|^2_{L^2(\Omega)} + \gamma|u(t)|^2\,dt + \alpha(|\omega|-c)^2,\quad \alpha>0,$$

$$A(\tau) := \det(\partial T_\tau)\partial T_\tau^{-1}\partial T_\tau^{-\top},\qquad \xi(\tau) := |\det(\partial T_\tau)|.$$

$$-\frac{d}{dt}\Pi_h = A_h\Pi_h+\Pi_h A_h-\Pi_h B_h\gamma^{-1}B_h^\top\Pi_h+M_h \quad \text{ in } [0,T),\quad \Pi_h(T)=0\,.$$

$$A(\tau,\cdot),\partial_\tau A(\tau,\cdot)\in L_\infty(\Omega,\mathbf{R}^{d\times d}),\quad \xi(\tau,\cdot),\partial_\tau\xi(\tau,\cdot)\in L_\infty(\Omega).$$

$$z_i[k] = \sum_{v_j\in\mathcal{N}_i^{\mathrm{mm}}[k]}\frac{1}{|\mathcal{S}_i^{\mathrm{mm}}[k]|}x_j[k]$$

$$[x_i[k+1]]_p = [z_i[k]]_p - \eta[k][g_i[k]]_p.$$

$$\|z_i[k]-\hat{x}\|_2 \le \sum_{v_j\in\mathcal{N}_i^{\mathrm{mm}}[k]}\frac{1}{|\mathcal{S}_i^{\mathrm{mm}}[k]|}\|x_j[k]-\hat{x}\|_2 \le R_i[k].$$

$$\min_{x\in\mathbb{R}^d}\frac{1}{N}\sum_{v_i\in\mathcal{V}} f_i(x)$$

$$D_{ij}[k]\triangleq \|x_j[k]-\hat{x}\|_2 \ \text{ for } \ x_j[k]\in\mathcal{S}_i[k]\setminus\{x_i[k]\}.$$

$\mathcal{C}_i(\epsilon) \triangleq \{x \in \mathbb{R}^d : f_i(x) \leq f_i(x_i^*) + \epsilon\}$.

$-\langle g(x),\ x^* - x\rangle \geq f(x) - f(x^*)$.

$\cos\hat{\theta}(x) \geq \dfrac{\epsilon}{L\delta(\epsilon)}$.

$\max\limits_{v_i \in \mathcal{R}} \|z_i[k] - \hat{x}\|_2 \leq \max\limits_{v_i \in \mathcal{R}} \|x_i[k] - \hat{x}\|_2$.

$\Gamma_i(p, l) \triangleq p^2 - \Delta_i(p, l)$.

$\Gamma_i(s^*(\xi),\ \eta[k]\ \|g_i[k]\|_2) \leq \big(s^*(\xi)\big)^2$.

$\Delta_i(\|z_i[k] - \hat{x}\|_2, \eta[k]\ \|g_i[k]\|_2) > \Delta_i(s^*(\xi), \eta[k]\ \|g_i[k]\|_2)$.

$\eta[k]\kappa_i \leq \eta[k]\ \|g_i[k]\|_2 \leq \eta[k]L \leq \dfrac{b_i}{2}$.

$\Delta_i\big(s^*(\xi),\ \eta[k]\ \|g_i[k]\|_2\big) \geq \dfrac{b_i}{2}\eta[k]\ \|g_i[k]\|_2$.

$\max\limits_{v_i \in \mathcal{R}} \|z_i[k^*] - \hat{x}\|_2 \leq \max\limits_{v_i \in \mathcal{R}} \|x_i[k^*] - \hat{x}\|_2 \leq s^*(\xi)$.

$\|g(x)\|_2\ \|x^* - x\|_2 \cos\hat{\theta}(x) \geq f(x) - f(x^*)$.

$\hat{\theta}(x) \leq \arccos\Big(\dfrac{\epsilon}{L\delta(\epsilon)}\Big) \triangleq \theta(\epsilon) < \dfrac{\pi}{2}$.

$$\begin{aligned}\Big\langle \sum_{v_i \in \mathcal{R}} g_i(x_0),\ u\Big\rangle &= \sum_{v_i \in \mathcal{R}} \langle g_i(x_0),\ u\rangle \\ &= \sum_{v_i \in \mathcal{R}} \|g_i(x_0)\|_2 \cos\angle(g_i(x_0),\ x_0 - \hat{x}) \\ &> 0\end{aligned}$$

$\max\limits_{y \in \mathcal{B}(\hat{x}, R)} \angle(x - y,\ x - \hat{x}) = \arcsin\Big(\dfrac{R}{\|x - \hat{x}\|_2}\Big)$.

$$\begin{aligned}\|x_i[k+1] - \hat{x}\|_2 &= \|z_i[k] - \hat{x} - \eta[k]g_i[k]\|_2 \\ &\leq \|z_i[k] - \hat{x}\|_2 + \eta[k]\|g_i[k]\|_2.\end{aligned}$$

$\eta[k_2^*] \leq \dfrac{1}{L}\min\limits_{v_i \in \mathcal{R}} \big\{\min\{a_i^+,\ b_i\}\big\}$.

$\eta[k]\ \|g_i[k]\|_2 \leq \eta[k]L \leq \min\limits_{v_j \in \mathcal{R}} \big\{\min\{a_j^+,\ b_j\}\big\} \leq a_i^+$.

$\mathcal{I}_x[k] \triangleq \{v_i \in \mathcal{R} : \|x_i[k] - \hat{x}\|_2 > s^*(\xi)\}$

$\mathcal{I}_z[k] \triangleq \{v_i \in \mathcal{R} : \|z_i[k] - \hat{x}\|_2 > s^*(\xi)\}$.

$$\max_{v_i \in \mathcal{I}_z[k]} \|x_i[k+1] - \hat{x}\|_2^2 \geq \max_{v_i \in \mathcal{I}_x[k+1]} \|x_i[k+1] - \hat{x}\|_2^2$$
$$= \max_{v_i \in \mathcal{R}} \|x_i[k+1] - \hat{x}\|_2^2.$$

$$\|z_i[k] - \hat{x}\|_2 \leq \max_{v_j \in (\mathcal{N}_i^- \cap \mathcal{R}) \cup \{v_i\}} \|x_j[k] - \hat{x}\|_2$$
$$\leq \max_{v_j \in \mathcal{R}} \|x_j[k] - \hat{x}\|_2,$$

$$\max_{v_i \in \mathcal{R}} \|x_i[k^* + K^*] - \hat{x}\|_2^2 \leq \big(s^*(\xi)\big)^2.$$

$$\|x_i[k^*] - \hat{x}\|_2 \leq s^*(\xi)$$

$$\max_{v_i \in \mathcal{R}} \|x_i[k^*] - \hat{x}\|_2 \leq s^*(\xi).$$

$$\max_{v_i \in \mathcal{R}} \|x_i[k^* + K] - \hat{x}\|_2 \leq s^*(\xi) \quad \text{for all } \; K \geq 0$$

$$\rho = \sum_{i=1}^{r} \rho_i^{(1)} \otimes \cdots \otimes \rho_i^{(n)} \quad \text{with all } \rho_i^{(j)} \succcurlyeq 0.$$

$$\operatorname{tr}((\sigma \otimes \tau) C_T) \geqslant 0 \quad \text{for all } \sigma \succcurlyeq 0, \tau \succcurlyeq 0.$$

$$\tau_i \in C_{t_i} \quad \text{with} \quad i = 1, \ldots, n,$$

$$v^* \tau v = \sum_{i,j} \bar{\lambda}_i \lambda_j \mu_{i,j}.$$

$$p = \sum_{\omega} \sigma_\omega \otimes \omega,$$

$$p = \sum_{i=1}^{d} E_{ii} \otimes z_i$$

$$(\tau_1 \oplus \chi_1, \ldots, \tau_d \oplus \chi_d) \in C_{r+s}(p),$$

$$\ell = \sigma_0 \otimes 1 + \sum_{i=1}^{d} \sigma_i \otimes z_i$$

$$(\tau_1, \tau_2) \in C_s(\ell).$$

$$0 \preccurlyeq \sigma_i \preccurlyeq I_m,$$

$$\ell := I_m \otimes 1 - \sum_{i=1}^{d} (2\sigma_i - I_m) \otimes z_i$$

$$v^* \sigma_i v = \tau_i \text{ for all } i = 1, \ldots, d.$$

$w\colon \mathcal{Q}_A \times \mathcal{Q}_B \times \mathcal{A}_A \times \mathcal{A}_B \to \{0,1\}$

$c_A\colon \mathcal{Q}_A \to \mathcal{A}_A \quad \text{and} \quad c_B\colon \mathcal{Q}_B \to \mathcal{A}_B,$

$p(x,y \mid a,b) = \sum_{\lambda} q_\lambda \, r_A(a,\lambda)(x) \cdot r_B(b,\lambda)(y),$

$\mathcal{G} := \mathbb{C}\langle \mathcal{Q}_A \times \mathcal{A}_A, \mathcal{Q}_B \times \mathcal{A}_B \rangle.$

$\varphi\colon \mathcal{G} \to \mathbb{C}$

$$\rho = \sum_{i_1,\ldots,i_n=1}^{r} \rho^{(1)}_{i_1,i_2} \otimes \rho^{(2)}_{i_2,i_3} \otimes \cdots \otimes \rho^{(n)}_{i_n,i_1}.$$

$$\rho = \xi\xi^* \quad \text{with} \quad \xi = \sum_{i_1,\ldots,i_n=1}^{r} \xi^{(1)}_{i_1,i_2} \otimes \xi^{(2)}_{i_2,i_3} \otimes \cdots \otimes \xi^{(n)}_{i_n,i_1}.$$

$$\rho = \sum_{i_1,\ldots,i_n=1}^{r} \rho_{i_1,i_2} \otimes \rho_{i_2,i_3} \otimes \cdots \otimes \rho_{i_n,i_1},$$

$$\rho = \sum_{\alpha} \rho_\alpha \otimes \rho_\alpha \quad \text{with all } \rho_\alpha \text{ psd},$$

$$\rho = \sum_{\alpha,\beta} \rho_{\alpha,\beta} \otimes \rho_{\beta,\alpha} \quad \text{with all } \rho_{\alpha,\beta} \text{ psd}.$$

$$\rho = \sum_{i} \rho^{(1)}_i \otimes \cdots \otimes \rho^{(n)}_i$$

$$\mathbb{P}\left(|X - \mathbb{E}X| > \frac{1}{2}\mathbb{E}X\right) \le \frac{4\mathrm{Var}(X)}{(\mathbb{E}X)^2} = 4\left(\frac{\mathbb{E}(X(X-1))}{(\mathbb{E}X)^2} + \frac{1}{\mathbb{E}X} - 1\right).$$

$$\mathbb{E}\ |S(v,H_i;t)| \le \frac{t^{w_i(v)}}{n^{w_i(v)-1}} \left(2(w_i(v)!)^D - 1\right),$$

$$\mathbb{E}\ |S(v,H_i;t)| \le \frac{t^{w}}{n^{w-1}} \left(2w^{d_i(x,v)}\left((w-1)!\right)^D - 1\right).$$

$$\tau_{\mathcal{P}}(q,n) = \inf_{\mathcal{S}} \tau_{\mathcal{P}}(\mathcal{S},q,n),$$

$$\mathbb{E}\ |S(v,H_i;t)| \le \frac{t^{w_i}}{n^{w_i-1}} \left(2\left((s-1)^{w_i} w_i!\right)^D - 1\right),$$

$$\begin{aligned}\mathbb{E}\ |S(v,H_i;t)| &\le \frac{t^{w}}{n^{w-1}} \left(2w^{d(x,u)+1}\left((w-1)!\right)^D - w + \frac{t}{n-t}\right) \\ &\le \frac{t^{w}}{n^{w-1}} \left(2w^{d(x,v)}\left((w-1)!\right)^D - 1\right),\end{aligned}$$

$$\mathbb{E}\ |S(v,H_i;t)| \leq (s-1)w_i\mathbb{E}\ |S(u,H_i;t)| + 2\sum_{a\geq w_i+1}(s-1)a\frac{t^a}{a!n^{a-1}}$$

$$\lim_{n\to\infty}\mathbb{P}[G_t \in \mathcal{P}_H] = 1.$$

$$\lim_{n\to\infty}\mathbb{P}[G_t \in \mathcal{P}_H] = 0.$$

$$\lim_{n\to\infty}\mathbb{P}[G_t^{(s)} \in \mathcal{P}_H] = 1.$$

$$\lim_{n\to\infty}\mathbb{P}[G_t^{(s)} \in \mathcal{P}_H] = 0.$$

$$X_t^{(x)} = \sum_{i=1}^{n} I_t(i)^{(x)},$$

$$\mathbb{E}(X(X-1)) = n(n-1)\mathbb{P}(I_t(1)^{(x)} = I_t(2)^{(x)} = 1),$$

$$\mathbb{P}(I_t(1)^{(x)} = I_t(2)^{(x)} = 1) = \binom{t}{x}\binom{t-x}{x}\left(\frac{1}{n}\right)^{2x}\left(1-\frac{2}{n}\right)^{t-2x} \sim \frac{t^{2x}}{x!^2n^{2x}},$$

$$w_{H_i}(x_i) = w_{H_{i-1}}(x_i) + 1$$

$$\mathbb{E}\ |S(v,H;t)| \leq \frac{t^w}{n^{w-1}}\left(2(m!)^D - 1\right)$$

$$S(v,H_0;t) = n = \frac{t^0}{n^{-1}}\left(2(0!)^D - 1\right),$$

$$|S(v,H_i;t)| \leq |S(v,H_{i-1};t)|,$$

$$\mathbb{E}\ |S(x,H_i;t)| \leq \sum_{t'=0}^{t-1}\frac{|S(x,H_{i-1};t')|}{n}$$

$$\mathbb{E}\ |S(v,H';t)| \leq \frac{t^d}{n^{d-1}}\left(2(e(H')!)^{|V(H')|-1} - 1\right) = O\left(\frac{t^d}{n^{d-1}}\right).$$

$$\mathbb{E}\ |S(v,H;t)| \leq \frac{t^w}{n^{w-1}}\left(2\left((s-1)^m m!\right)^D - 1\right)$$

$$\tau_{\mathcal{P}} := \lim_{q\to 1^-}\limsup_{n\to\infty}\frac{\ln(\tau_{\mathcal{P}}(q,n))}{\ln n}$$

$$[X_1] = [X_2] \iff [X_1 \cdot X_2^{-1}] = 0.$$

$$[C] = [0] \iff [\sigma_I(C)] = [0].$$

$$[X_1 \cdot X_2 \cdot X_3 \cdot X_4] = [0] \iff [X_2 \cdot X_3 \cdot X_4 \cdot X_1] = [0]$$

$$[X_1^{-1}] = [X_2 \cdot X_3 \cdot X_4] \iff [X_2^{-1}] = [X_3 \cdot X_4 \cdot X_1].$$

$$[\phi(C)] = [\beta_{W_v}(\psi(C))].$$

$$[\phi(U)] = [W_u \cdot \psi(U) \cdot W_v^{-1}].$$

$$[C] = [C'] \text{ implies } [\![C]\!] = [\![C']\!].$$

$$(r)(r, x_1)(r, x_1, x_2) \dots (r, x_1, x_2 \dots, x_\ell).$$

$$\phi : \mathcal{U} \to H : (r, x_1, \dots, x_\ell) \mapsto x_\ell$$

$$\Phi : (r, x_1, \dots, x_\ell) \mapsto (\phi(r), \phi(r, x_1), \dots, \phi(r, x_1, \dots, x_\ell))$$

$$\Phi^{-1}(X)_i := \Phi^{-1}(X)_{i-1} \cdot \phi_{x_{i-1}}^{-1}(x_i)$$

$$W_d = \sqrt{A}^d \cdot T \cdot I \cdot S^{-1}$$

$$[\sqrt{A}^d \cdot T \cdot I \cdot S^{-1}] = [\sqrt{B}^{d'} \cdot T' \cdot I' \cdot S'^{-1}]$$

$$[T \cdot \sqrt{A_f}^d \cdot I \cdot S^{-1}] = [T' \cdot \sqrt{B}_f^{d'} \cdot I' \cdot S'^{-1}]$$

$$[\phi(C)] \neq [\beta_W(\psi(C))]$$

$$W_d := \sqrt{\phi(C)}^d \cdot T \cdot I \cdot S^{-1}$$

$$[W_u] = [\phi(e) \cdot W_v \cdot \phi(e)^{-1}],$$

$$g(t) := \frac{1}{c + (k-1)t}$$

$$\begin{aligned} H &= \sum_{y=1}^{g} \prod_{i=1}^{k} (1 - \chi_{D'}(D'y + n' + h_i)\chi_g(D'y + n' + h_i)) \\ &= \sum_{y=1}^{g} \prod_{i=1}^{k} (1 - \chi_{D'}(n' + h_i)\chi_g(D'y + n' + h_i)) \end{aligned}$$

$$H = \left| \left(\sum_{y=1}^{g} \prod_{i=1}^{k} \epsilon_i \right) + \left(\sum_{y=1}^{g} \sum_{j=1}^{2^k - 1} \left(\prod_{i \in S_j} \epsilon_i \right) \chi_g(Q_j(y)) \right) \right|$$

$$\begin{aligned} m_2 &:= \int_0^T g(t)^2 \, dt \\ \mu &:= \frac{1}{m_2} \int_0^T t g(t)^2 \, dt \\ \sigma^2 &:= \frac{1}{m_2} \int_0^T t^2 g(t)^2 \, dt - \mu^2. \end{aligned}$$

$$\pi(x,q,a) = \frac{\pi(x)}{\phi(q)}\left(1 - \chi_D\left(\frac{aD}{(q,D)}\right) + O\left(L(1,\chi_D)(\log x)^{r^r}\right)\right),$$

$$I_k(F) = \int_0^\infty ... \int_0^\infty F(t_1,...,t_k)^2 dt_1 ... dt_k,$$
$$J_k^{(m)}(F) = \int_0^\infty ... \left(\int_0^\infty F(t_1,...,t_k)^2 dt_m\right) dt_1 ... dt_{m-1} dt_{m+1} dt_k.$$

$$\left(\int_0^{r-X_1-\cdots-X_{k-1}} g(t)^2 dt\right) \leq \frac{m_2 M_k^{[T]} r}{k} \mathbb{P}\left(X_1 + \cdots + X_k \geq r\right)$$

$$\begin{aligned} H_1 &\leq 12, \\ H_2 &\leq 270, \\ H_3 &\leq 52,116, \\ H_4 &\leq 474,266, \\ H_5 &\leq 4,137,854, \\ H_m &\ll me^{2m} \end{aligned}$$

$$\begin{aligned} H_1 &= 2 \\ H_2 &\leq 12, \\ H_4 &\leq 270, \\ H_6 &\leq 52,116, \\ H_m &\ll e^{(1+\epsilon')m}, \end{aligned}$$

$$\begin{aligned} 5.9484 &> \frac{3}{.504346916} = 5.94828... \\ 7.9310494 &> \frac{4}{.504346916} = 7.9310487... \\ 9.9138119 &> \frac{5}{.504346916} = 9.9138109... \end{aligned}$$

$H_1 \leq 70,000,000,$

$$L(s,\chi_q) = \sum_{n=1}^{\infty} \frac{\chi_q(n)}{n^s}.$$

$\beta < 1 - C(\epsilon) q^{-\epsilon}.$

$$s_D > 1 - \frac{1}{(\log D)^{r^r+A}}.$$

$q < x^{\frac{58\left(1-\frac{1}{r}\right)}{115}} = x^{.5043469...},$

$$M_k > \frac{2m}{\theta'}.$$

$T = \{n + h_1, \cdots, n + h_k\}.$

$$Re(s) > 1 - \frac{1}{(\log D)^{r^r+A}} > 1 - \frac{1}{(\log C)^{r^r+A}}.$$

$$\prod_{i=1}^{k} (1 - \chi_D(x + n' + h_i)).$$

$$\left| \sum_{y=1}^{g} \chi_g(Q_j(y)) \right| \leq (k-1)\sqrt{g}.$$

$$l \equiv D'y' + n' \pmod{D}$$

$$\mathcal{R}_k = \{(t_1, ..., t_k) : \sum_{i=1}^{k} t_i \leq 1\}$$

$$B = \frac{\phi(W)}{W} \log R,$$

$$S_2 = (\theta + o(1)) \frac{X}{W} B^{-k} \sum_{m=1}^{k} J_k^{(m)}(F).$$

$$S = \left((\theta + o(1)) \sum_{m=1}^{k} J_k^{(m)}(F) - \rho I_k(F) \right) \frac{X}{W} B^{-k}.$$

$$\pi(x, q, a) = \frac{2\pi(X)}{\phi(q)} + O\left(\frac{\pi(X)}{\phi(q) \log^{A-2}(x)} \right).$$

$$(\theta - 2\epsilon) M_k > \rho.$$

$$M_k \geq \log k - 2 \log\log k - 2.$$

$$k \gg e^{(1.98276)\rho}.$$

$$\mathcal{R}_k^{[\alpha]} = \{(t_1, ..., t_k) \in [0, \alpha]^k : \sum_{i=1}^{k} t_i \leq 1\}.$$

$$p \mapsto \langle V(p), W(p) \rangle_p$$

$$F(p) = 0_p \in T_p\mathcal{M}.$$

$$\operatorname{grad}_x \mathcal{L}(x, y, z) = 0_x;\ h(x) = 0, g(x) \leq 0, z \geq 0;\ Zg(x) = 0.$$

$$\bar{\mathrm{R}}_w(\xi) := (\mathrm{R}_x(\xi_x), y + \xi_y, z + \xi_z, s + \xi_s)$$

$$\operatorname{Hess}_x \mathcal{L}(w) = \operatorname{Hess} f(x) + \sum_{i=1}^{l} y_i \operatorname{Hess} h_i(x) + \sum_{i=1}^{m} z_i \operatorname{Hess} g_i(x),$$

$$F_\mu(w) := F(w) - \mu\hat{e},\ \text{and}\ \hat{e} := \hat{e}(w) := (0_x, 0, 0, e).$$

$$\nabla F_{\mu}(w) = \nabla F(w) - \mu \nabla \hat{e}(w) = \nabla F(w),$$

$$\nabla F(w_k)[\Delta w_k] = -F(w_k) + \mu_k \hat{e};$$

$$\mathcal{T}(\Delta x, \Delta y) := \begin{pmatrix} \mathcal{A}_w[\Delta x] + \mathcal{H}_x[\Delta y] \\ \mathcal{H}_x^*[\Delta x] \end{pmatrix} = \begin{pmatrix} c \\ q \end{pmatrix}.$$

$$\mathcal{A}_w[\Delta x] = c$$

$$\mathrm{T}_{\eta}(\xi) := \mathrm{P}_{\gamma}^{1 \to 0}(\xi)$$

$$\nabla F(p_k)[\xi_k] = -F(p_k) + \mu_k \hat{e};$$

$$\|F(p_k)\| = O(d(p_k, p^*)).$$

$$\alpha^i := \max_{\alpha \in (0,1]} \left\{ \alpha : f^i(t) \geq 0, \text{ for all } t \in (0, \alpha] \right\},$$

$$\nabla F(w_k)[\Delta w_k] = -F(w_k) + \sigma_k \rho_k \hat{e};$$

$$y = \left[\left(\mathcal{H}_x^* \mathcal{H}_x \right)^{-1} \mathcal{H}_x^* \right] \left(\operatorname{grad}_x \mathcal{L}(w) - \operatorname{grad} f(x) - \mathcal{G}_x z \right).$$

$$\varphi'(\alpha) = \mathcal{D}\varphi(\bar{\mathrm{Exp}}_{w_k}(\alpha \Delta w_k))[\mathcal{D}\bar{\mathrm{Exp}}_{w_k}(\alpha \Delta w_k)[\Delta w_k]].$$

$$\operatorname{grad}_x \mathcal{L}(x, y, z) = \operatorname{grad} f(x) + \sum_{i=1}^{l} y_i \operatorname{grad} h_i(x) + \sum_{i=1}^{m} z_i \operatorname{grad} g_i(x),$$

$$\left\| \mathrm{P}_{\gamma}^{0 \to 1}[F(y)] - F(x) - \int_0^1 \mathrm{P}_{\gamma}^{0 \to t} \nabla F(\gamma(t)) \mathrm{P}_{\gamma}^{t \to 0}[\eta] \mathrm{d}t \right\| \leq c_1 \|\eta\|^2,$$

$$p_{k+1} := \mathrm{R}_{p_k}[\alpha_k \nabla F(p_k)^{-1}(-F(p_k) + \mu_k \hat{e})],$$

$$\Pi := 2 \max \left\{ \max_i \left\{ 1/(s^*)_i \mid (s^*)_i > 0 \right\}, \max_i \left\{ 1/(z^*)_i \mid (z^*)_i > 0 \right\} \right\}.$$

$$\tau_1 := \min(Z_0 S_0 e)/(z_0^T s_0/m), \ \tau_2 := z_0^T s_0 / \|F(w_0)\|.$$

$$\langle \operatorname{grad} \varphi(w_k), \Delta w_k \rangle = 2(-\|F(w_k)\|^2 + \sigma_k \rho_k z_k^T s_k).$$

$$\|\mathcal{A}_x y\|_x^2 = \| \sum_{i=1}^{n} (\hat{\mathcal{A}}_x \hat{y})_i E_i \|_x^2 = \sum_{i=1}^{n} (\hat{\mathcal{A}}_x \hat{y})_i^2 = \|\hat{\mathcal{A}}_x \hat{y}\|_2^2.$$

$$\hat{\nabla} F(w_k) = \begin{pmatrix} Q_k & B_k & C_k & 0 \\ B_k^T & 0 & 0 & 0 \\ C_k^T & 0 & 0 & I \\ 0 & 0 & S_k & Z_k \end{pmatrix}.$$

$$\min_{X \in \mathbb{R}^{n \times k}} -2 \operatorname{trace}(X^T C) \quad \text{s.t. } X^T X = I_k, \ X \geq 0.$$

$$\min_{X\in\mathbb{R}^{m\times n}} \|A-X\|_{\mathrm{F}}^2 \quad \text{s.t. } \operatorname{rank}(X)=r,\ X\geq 0.$$

$$d(p_{k+1},p^*)=(1-\alpha_k)O(d(p_k,p^*))+O(d^2(p_k,p^*))+O(\mu_k).$$

$$S_k^{-1}\Delta s_k+Z_k^{-1}\Delta z_k=\mu_k(S_kZ_k)^{-1}e-e,$$

$$\|Q(w)\|_{\mathrm{F}}\leq\|\hat{\mathrm{Hess}}f(x)\|_{\mathrm{F}}+\sum_{i=1}^{l}|y_i|\|\hat{\mathrm{Hess}}h_i(x)\|_{\mathrm{F}}+\sum_{i=1}^{m}|z_i|\|\hat{\mathrm{Hess}}g_i(x)\|_{\mathrm{F}}=:T(w).$$

$$\lim_{t\to s+}\|u(t,\cdot;s,u_0)-u_0(\cdot)\|_{C^0(\bar{\Omega})}=0,$$

$$\limsup_{t\to\infty}\int_\Omega u^{-p}(t,x;s,u_0)dx<\infty$$

$$\lim_{t-s\to\infty}\int_\Omega u^{-p}(t,x;s,u_0)\leq M_1(p).$$

$$\|u(t,\cdot;s,u_0)\|_\infty\leq M_4^*(\theta)\quad\forall t-s\gg 1,$$

$$\int_\Omega\tilde{u}_0(x)dx\geq\frac{2}{3}\int_\Omega u_0(x)dx\quad\text{for any }\tilde{u}_0\in\mathcal{B}(u_0,r).$$

$$\int_\Omega u^{-1}(t,x;s,u_0)dx\leq\max\Big\{\int_\Omega u^{-1}(\tau,x;s,u_0)dx,\frac{b_{\sup}|\Omega|}{a_{\inf}-a_{\chi,\mu}}\Big\}$$

$$\int_\Omega u^{-p}(t,x;s,u_0)dx\leq M_1^*\quad\forall\ s<t<\infty$$

$$\int_\Omega u^{q}(t,x;s,u_0)dx\leq M_2^*\quad\forall\, t\geq s.$$

$$\liminf_{t-s\to\infty}\inf_{x\in\Omega}u(t,x;s,u_n,a,b)\leq m_n:=\frac{1}{n}.$$

$$u_1^*(x)=u(1,x;0,u_0^*,a^*,b^*).$$

$$u(\cdot,t;0,u^*)=u^*(\cdot)\quad\text{for all }t\geq 0.$$

$$\int_\Omega\frac{|\nabla v(x,t;s,u_0)|^{2q}}{v^{k+1}(x,t;s,u_0)}\leq M(k,q)\int_\Omega\frac{u^q(x,t;s,u_0)}{v^{k-q+1}(x,t;s,u_0)}+M^*(k,q)\int_\Omega v^{2q-k-1}(x,t;s,u_0)$$

$$\int_\Omega u^q(t,x;s,u_0)dx\leq\max\left\{\int_\Omega u^q(\tau,x;s,u_0)dx,M_2(p,q)+\tilde{M}_2(p,q,\tau,s,u_0)\right\}\quad\forall\,t\geq\tau,$$

$$\|A_p^\beta e^{-A_pt}\|_{L^p(\Omega)}\leq C_{p,\beta}t^{-\beta}e^{-\gamma t}\ \text{ for }\ t>0.$$

$$(\nu A^{-1}\mathcal{M}(\tilde{u}_0,a,b)u)(t,x)\geq\delta_0\int_\Omega\mathcal{M}(\tilde{u}_0,a,b)u(x,t)dx\geq\frac{\delta_0}{2}\int_\Omega u_0(x)dx=\varepsilon$$

$$\frac{(p+1)\beta\mu}{p} - R < 0,$$

$$\frac{(p+1)\beta\mu}{p} = 2\beta\mu = 2(\chi + 2 - 2\sqrt{\chi+1})\mu = a_{\chi,\mu} < a_{\inf}.$$

$$\frac{b_{\sup}|\Omega|(q-1)\Big\{(C^*(n,\nu))^{\frac{1}{q+1}} + \varepsilon\Big\}\max\{\chi,\chi^2\}}{4\delta_0(a_{\inf} - a_{\chi,\mu})} < b_{\inf}.$$

$$|\Omega_\epsilon| = \int_\Omega u_n^{-p}\cdot u_n^p \le \int_\Omega u_n^{-p}\cdot(2\epsilon)^p \le (2\epsilon)^p \int_\Omega u_n^{-p} \le (2\epsilon)^p M_1^*.$$

$$u(t+T,\cdot;0,u^T) = u(t,\cdot;T,u(T,\cdot;0,u^T)) = u(t,\cdot;0,u^T).$$

$$\mathcal{Y}(T) = \{a(\cdot,\cdot)\in\mathcal{X}_T\,:\,|a(t,x)-a(t^{'},x)|\le B_1|t-t^{'}|^\gamma,\ \alpha\le a(t,x)\le B_2\ \forall\, t,t^{'}\in[s,s+T],\ x\in\bar{\Omega}\},$$

$$\int_\Omega g(\lambda u_1(x)+(1-\lambda)u_2(x))dx \le \lambda\int_\Omega g(u_1(x))dx + (1-\lambda)\int_\Omega g(u_2(x))dx.$$

$$0 = \Delta v - \frac{1}{|\Omega|}\int_\Omega u(\cdot,t) + u, \quad x\in\Omega$$

$$(u(t,x;s,u_0,a,b),v(t,x;s,u_0,a,b)) = (u(t,x;s,u_0),v(t,x;s,u_0)).$$

$$\|u\|_{\mathcal{X}_T} = \max_{s\le t\le s+T}\|u(t,x)\|_{C^0(\bar{\Omega})}.$$

$$\lim_{t-s\to\infty}\int_\Omega u^{-1}(t,x;s,u_0)dx = \frac{b_{\sup}|\Omega|}{a_{\inf}-a_{\chi,\mu}}.$$

$$2\beta - \frac{2N}{q} > \theta.$$

$$\lim_{n\to\infty} a_n(t,x) = a^*(t,x), \quad \lim_{n\to\infty} b_n(t,x) = b^*(t,x)$$

$$\liminf_{t-s\to\infty}\inf_{x\in\Omega} u(t,x;s,u_0,a,b) \ge m^*.$$

$$0\le u_n(x)\le 2\varepsilon \quad \forall\, x\in\Omega_\varepsilon.$$

$$\mathcal{L}_{QED} = \bar{\psi}(i\ \not{\partial} - m)\psi - \bar{\psi}\psi\gamma_5\psi - e\bar{\psi}A\psi$$

$$S_{eff}[b,A] = -i\,\mathrm{Tr}\ln(\not{p} - m - \not{\beta}\gamma_5 - e\ \not{A})$$

$$S'_{eff}[b,A] = i\,\mathrm{Tr}\sum_{n=1}^{\infty}\frac{1}{n}\left[\frac{1}{\not{p}-m-\not{p}\gamma_5}e\ \not{A}\right]^n.$$

$$S'_{eff}[b,A] = S^{(2)}_{eff}[b,A] + \ldots$$

$$S_{\rm eff}^{(2)}[b,A] = \frac{ie^2}{2}\,{\rm Tr}\left[\frac{1}{\not p - m - \not b\gamma_5} A \infty \frac{1}{\not - m - \not p\gamma_5} A\right].$$

$$S_{\rm eff}^{(2)}[b,A(x)] = \frac{1}{2}\int d^4x \Pi^{\alpha\mu\nu} F_{\alpha\mu} A_\nu$$

$$\Pi^{\alpha\mu\nu} = -\frac{ie^2}{2}\int \frac{d^4p}{(2\pi)^4}\,{\rm tr}\left[S_b(p)\gamma^\mu S_b(p)\gamma^\alpha S_b(p)\gamma^\nu\right],$$

$$S_b(p) = \frac{i}{\not p - m - \not p\gamma_5}$$

$$S_b(p) = i\left[\frac{\not p + m - \gamma_5\ \not p}{(p^2-m^2)} - \frac{2\gamma_5(m\ \not p - (b\cdot p))(\not p + m)}{(p^2-m^2)^2}\right] + \cdots.$$

$$p_\rho \to \vec{p}_\rho + p_0\delta_{0\rho}$$

$$S_{\rm eff}^{\rm cov} = \frac{(-i)}{2}\int d^4x I_1 \varepsilon^{\alpha\mu\nu\beta} b_\beta F_{\alpha\mu} A_\nu$$

$$I_1 = \frac{-3ie^2}{2\pi}\int \frac{d^D\vec{p}}{(2\pi)^D}\int_{-\infty}^{+\infty} dp_0 \left[\frac{\left(1-\frac{4}{D}\right)\vec{p}^2 + p_0^2 - m^2}{\left(\vec{p}^2 + p_0^2 + m^2\right)^3}\right]$$

$$I_2 = \frac{ie^2}{\pi}\int \frac{d^D\vec{p}}{(2\pi)^D}\int_{-\infty}^{+\infty} dp_0 \left[\frac{\vec{p}^2 - p_0^2}{\left(\vec{p}^2 + p_0^2 + m^2\right)^3}\right].$$

$$I_1 = \frac{3ie^2}{8D}\int \frac{d^D\vec{p}}{(2\pi)^D}\left[\frac{2(3-D)\vec{p}^2 + Dm^2}{\left(\vec{p}^2+m^2\right)^{5/2}}\right]$$

$$I_2 = -\frac{ie^2}{8D}\int \frac{d^D\vec{p}}{(2\pi)^D}\left[\frac{(3-D)\vec{p}^2 - Dm^2}{\left(\vec{p}^2+m^2\right)^{5/2}}\right]$$

$$I_1 = \frac{3ie^2}{8(4\pi)^{D/2}}\frac{\Gamma\left(1+\frac{\epsilon}{2}\right)}{\Gamma\left(\frac{5}{2}\right)\left(m^2\right)^{\epsilon/2}}$$

$$S_{\rm eff}^{\rm cov} = \frac{1}{2}\int d^4x \varepsilon^{\alpha\mu\nu\beta} k_\beta F_{\alpha\mu} A_\nu$$

$$k_\beta = \frac{e^2}{16\pi^2} b_\beta$$

$$S_b(p) = \frac{i}{\not p - m} + \frac{i}{\not p - m}\left(-ip\gamma\gamma_5\right)\frac{i}{\not p - m} + \cdots$$

$$\begin{aligned}\Pi_{\mathbf{e}}^{\mu\alpha\nu} =& -2ie^2\epsilon^{\mu\alpha\nu\rho}\int \frac{d^4p}{(2\pi)^4}\frac{1}{\left(p^2+m^2\right)^3}\\ &\times\left[b_\rho\left(p^2-3m^2\right) - 4p_\rho(b\cdot p)\right],\end{aligned}$$

$$\tilde{k}_\beta = \frac{e^2}{4\pi^2} b_\beta .$$

$$\tilde{k}_\beta = k_\beta + \frac{3e^2}{16\pi^2} b_\beta$$

$$\Delta k_\beta = \frac{3e^2}{16\pi^2} b_\beta$$

$$S'_{eff}[b,A] = i \,\mathrm{Tr} \sum_{n=1}^{\infty} \frac{1}{n} \left[\frac{1}{\not{p} - m - e \not{A}} \not{\gamma}_5 \right]^n .$$

$$S^{(1)}_{eff}[b,A] = i \,\mathrm{Tr} \left[\frac{1}{\not{p} - m - e\not{A}} \not{\gamma}_5 \right] .$$

$$\frac{1}{A-B} = \frac{1}{A} + \frac{1}{A} B \frac{1}{A} + \frac{1}{A} B \frac{1}{A} B \frac{1}{A} + \cdots$$

$$S^{(1)}_{eff}[b,A] = ie^2 \,\mathrm{Tr} \left[\frac{1}{p-m} \not{\gamma}_5 \frac{1}{p-m} A \frac{1}{p-m} A \right] ,$$

$$S^{(1)}_{\mathrm{eff}}[b,A(x)] = \frac{1}{2} \int d^4x \Pi^{\alpha\mu\nu} F_{\alpha\mu} A_\nu ,$$

$$\int \frac{d^4p}{(2\pi)^4} p_\mu p_\nu f\left(p^2\right) = \frac{g_{\mu\nu}}{4} \int \frac{d^4p}{(2\pi)^4} p^2 f\left(p^2\right) ,$$

$$\Pi^{\mu\alpha\nu} = \frac{e^2}{16\pi^2} \varepsilon^{\mu\alpha\nu\theta} b_\theta$$

$$\rho^2 = \rho \, .$$

$$\mathrm{Tr}\, \rho \, = \, \sum_k \mathbb{P}(k) \, = \, 1 \, ,$$

$$\frac{d\rho}{dt} \, = \, -i \left(\mathrm{H}_{\mathrm{eff}} \, \rho - \rho \, \mathrm{H}^\dagger_{\mathrm{eff}} \right) \, = \, -i \left[\mathrm{E}, \rho \right] - \frac{1}{2} \left\{ \Gamma, \rho \right\} .$$

$$\frac{d}{dt} \mathrm{Tr}\, \rho \, = \, \frac{d}{dt} \sum_k \mathbb{P}(k) \, = \, -\mathrm{Tr}\big(\Gamma\rho\big) \, \neq \, 0 \, .$$

$$\mathbb{P}(k|\mathcal{H}_1) \, = \, \frac{\mathbb{P}(k)}{\mathbb{P}(\mathcal{H}_1)} \, = \, \frac{\mathbb{P}(k)}{\sum_{q\in\mathcal{H}_1} \mathbb{P}(q)} \, = \, \frac{\mathbb{P}(k)}{\mathrm{Tr}\, \rho} \, ,$$

$$\rho = \frac{1}{2} a_\mu \overline{\sigma}^\mu , \qquad E = E_\mu \sigma^\mu , \qquad \Gamma = \Gamma_\mu \sigma^\mu ,$$

$$\frac{da^0}{dt} = -\Gamma_\mu a^\mu ,$$

$$\tau \,=\, |\mathbf{\Gamma}|\, t, \qquad r \,=\, \frac{|\mathbf{\Gamma}|}{2\,|\mathbf{E}|}\,,$$

$$\frac{d\mathbf{b}}{d\tau} \,=\, -\frac{1}{r}(\mathbf{e}\times\mathbf{b}) \,+\, \boldsymbol{\gamma} \,-\, (\boldsymbol{\gamma}\cdot\mathbf{b})\,\mathbf{b}\,.$$

$$\cos\theta_{\mathbf{b}_\star\boldsymbol{\gamma}} \,=\, \boldsymbol{\gamma}\cdot\mathbf{b}_\star \,=\, \frac{c_\gamma}{\alpha}\,.$$

$$\lim_{r\to 0}\theta_{\mathbf{b}_\star\boldsymbol{\gamma}} = \theta_{\mathbf{e}\boldsymbol{\gamma}}, \qquad \lim_{r\to\infty}\theta_{\mathbf{b}_\star\boldsymbol{\gamma}} = 0\,.$$

$$X \,=\, \begin{pmatrix} p & p \\ -q & q \end{pmatrix}\,,$$

$$\frac{d\rho}{dt} \,=\, -i\,[\mathrm{E},\rho] \,-\, \frac{1}{2}\,\{\Gamma,\rho\} \,-\, [\mathrm{D},[\mathrm{D},\rho]]\,,$$

$$\frac{d|\mathbf{b}|^2}{d\tau} \,=\, 2\big(1-|\mathbf{b}|^2\big)\,|\mathbf{b}|\cos\theta_{\mathbf{b}\boldsymbol{\gamma}} \,-\, 2\zeta|\mathbf{b}|^2\sin^2\theta_{\mathbf{b}\mathbf{d}}\,.$$

$$\mathbf{b}_\star \;\to\; \mathbf{b}_\star + \zeta\mathbb{J}^{-1}\mathbf{b}_\star - \zeta(\mathbf{d}\cdot\mathbf{b}_\star)\mathbb{J}^{-1}\mathbf{d}\,.$$

$$S[\rho] \,=\, -\,a^0\ln a^0 \,+\, a^0\,S[\hat\rho] \,=\, -\,a^0\ln a^0 \,-\, a^0\,\mathrm{Tr}\big(\hat\rho\ln\hat\rho\big)\,,$$

$$S[\hat\rho] \,=\, \ln 2 - \frac{1}{2}\,(1-|\mathbf{b}|)\ln(1-|\mathbf{b}|) - \frac{1}{2}\,(1+|\mathbf{b}|)\ln(1+|\mathbf{b}|)\,.$$

$$S[\hat\rho_{\mathrm{Coh}}] \,=\, 0\,, \qquad S[\hat\rho_{\mathrm{Mix}}] \,=\, \ln 2\,,$$

$$\mathcal{T} \;\equiv\; \frac{(\mathbf{b}'\times\mathbf{b}'')\cdot\mathbf{b}'''}{|\mathbf{b}'\times\mathbf{b}''|^2} \;=\; 0,$$

$$\boldsymbol{\gamma} = \cos\varphi\,\hat{\mathbf{R}} - \sin\varphi\,\hat{\boldsymbol{\varphi}}, \qquad \mathbf{e}\times\boldsymbol{\gamma} = \sin\varphi\,\hat{\mathbf{R}} + \cos\varphi\,\hat{\boldsymbol{\varphi}}, \qquad \mathbf{e} = \hat{\mathbf{z}}.$$

$$\frac{d\varphi}{d\tau} \;=\; -\,\frac{1}{r} \,-\, \frac{\sin\varphi}{|\mathbf{b}|}\,.$$

$$|\mathbf{b}(\tau)| \;=\; \sqrt{1 \,-\, \mathcal{A}\,e^{2r\sin\varphi(\tau)}}\,,$$

$$\widehat{\mathrm{P}} \;\simeq\; \frac{2\pi r}{1-\mathcal{A}}\left\{\left[1-\left(\frac{2-2\mathcal{A}}{2-\mathcal{A}}\right)^2 r^2\right]^{-1/2} - \,\mathcal{A}\right\}.$$

$$|\mathbf{b}| \,=\, 1\,, \qquad \widehat{\mathrm{P}} \,=\, \frac{2\pi r}{\sqrt{1-r^2}}\,.$$

$$\mathbf{b}_\star = \left(\sqrt{1-\frac{1}{r^2}+\frac{\zeta^2}{4}} - \frac{\zeta}{2}\right)\boldsymbol{\gamma} - \frac{1}{r}\mathbf{e}\times\boldsymbol{\gamma}.$$

$$\mathbf{b}_\star \;=\; -\,r\,\mathbf{e}\times\boldsymbol{\gamma}\,.$$

$$\mathbf{n}_1 = \mathbf{e}, \qquad \mathbf{n}_2 = \frac{1}{s_\gamma}\mathbf{e}\times\boldsymbol{\gamma}, \qquad \mathbf{n}_3 = \frac{1}{s_\gamma}\mathbf{e}\times(\mathbf{e}\times\boldsymbol{\gamma}),$$

$$\mathbf{b} = \alpha\mathbf{n}_1 + \beta\mathbf{n}_2 + \eta\mathbf{n}_3,$$

$$\alpha^4 - (1-r^2)\alpha^2 - c_\gamma^2 r^2 = 0\,.$$

$$\alpha \;=\; \pm\frac{1}{\sqrt{2}}\sqrt{1-r^2+\sqrt{(1-r^2)^2+4c_\gamma^2 r^2}}\,.$$

$$S_{\mathrm{AHM}}[\phi, A_\mu] = \int \mathrm{d}^3x \left[-\frac{1}{4}F_{\mu\nu}^2 + |D_\mu\phi|^2 - \frac{\mu^2}{8}\left(|\phi|^2 - v^2\right)^2\right],$$

$$\phi_\infty : S^1_\infty \to U(1) \cong S^1,$$

$$m_\sigma = m_A \equiv m.$$

$$\rho(r) = \left(v^2 - \frac{1}{e}\frac{A'(r)}{r}\right)^{1/2},$$

$$\left(\frac{d^2}{d\xi^2} - \frac{1}{\xi}\frac{d}{d\xi} - 1\right)G(\xi,\xi') = \delta\left(\xi - \xi'\right),$$

$$Z_N = N + \frac{\pi}{3\sqrt{3}}N^2 + \frac{\pi^2}{108}N^3 + \mathcal{O}\left(N^4\right).$$

$$R_{\mathrm{core}} \equiv \sqrt{2\times\frac{\int \mathrm{d}^2\mathbf{x}|\mathbf{x}|^2\mathcal{H}(\mathbf{x})}{\int \mathrm{d}^2\mathbf{x}\mathcal{H}(\mathbf{x})}},$$

$$R_{\mathrm{core}} \ll R_{\mathrm{interaction}},$$

$$S_{\mathrm{REFT}}\left[\Phi,\sigma,A_\mu\right] = S_{\mathrm{AHM}}\left[\sigma,A_\mu\right] + S_{\mathrm{vortex}}\left[\Phi,\sigma,A_\mu\right].$$

$$\overline{\Sigma} = \tilde{\Psi}^+ + \eta\overline{\Phi}.$$

$$c_{\Gamma\Gamma\Gamma} = \sqrt{\pi}M^{-1/2}N^{1/2}.$$

$$M \sim \mathbf{p} \sim \mathfrak{m} \gg \mathbf{q} \sim m.$$

$$\int \frac{\mathrm{d}^{d-1}\mathbf{l}}{(2\pi)^{d-1}}\frac{\mathrm{d}\omega}{2\pi}\frac{\mathcal{N}(\omega,\mathbf{l})}{[\omega^2 - \mathbf{l}^2 - m^2 + i0][\omega^2 - (\mathbf{l}+\mathbf{q})^2 - m^2 + i0]},$$

$$H_{\mathrm{NREFT}}\left(\mathbf{p}_1,\mathbf{p}_2,\mathbf{x}_1,\mathbf{x}_2\right) = \sum_{i=1}^{2}\sqrt{\mathbf{p}_i^2 + M^2} + V\left(\mathbf{p}_1,\mathbf{p}_2,\mathbf{x}_1,\mathbf{x}_2\right).$$

$$V\left(\mathbf{p},\mathbf{p}'\right) = \sum_{i=1}^{\infty} V^{(i)}\left(\mathbf{p},\mathbf{p}'\right),$$

$$(\omega, \mathbf{l}) \sim (|\mathbf{p}||\mathbf{q}|, |\mathbf{p}|)\,.$$

$$\mathcal{M}^{(1)} = -\frac{32\pi MN\left(M^2 - (p_1 \cdot p_2)\right)}{q^2 - m^2} + \frac{8\pi MNm^2}{q^2 - m^2} + \frac{32\pi MN}{m^2}\left(p_1 \cdot p_2 + \frac{1}{4}\left(q^2 + m^2\right)\right).$$

$$V^{(1)}\left(\mathbf{p}, \mathbf{x}\right) = \frac{8MN\mathbf{p}^2}{\mathbf{p}^2 + M^2} K_0\left(mr\right).$$

$$g_{\sigma\sigma} = -2\pi.$$

$$L(\dot{\mathbf{x}}_1, \mathbf{x}_1; \dot{\mathbf{x}}_2, \mathbf{x}_2) = \frac{1}{2}M\dot{\mathbf{x}}_1^2 + \frac{1}{2}M\dot{\mathbf{x}}_2^2 - \tilde{U}\left(r_{12}\right)|\dot{\mathbf{x}}_1 - \dot{\mathbf{x}}_2|^2,$$

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

$$W = -\frac{1}{2}\epsilon_{\mu\nu\rho}P^{\mu}M^{\nu\rho}.$$

$$\varepsilon_{\pm}^{\mu}(-p) = -\varepsilon_{\mp}^{\mu}(p),$$

$$-\int \frac{\mathrm{d}^d l}{(2\pi)^d} \frac{1}{[l^2 - m^2][(l+q)^2 - m^2][p_1 \cdot l - i0]}.$$

$$\int \frac{\mathrm{d}^2\mathbf{l}}{(2\pi)^2}\, \mathbf{u} \cdot \frac{\partial}{\partial \mathbf{l}}\left(\frac{(\mathbf{u}\cdot\mathbf{l})^n}{[\mathbf{l}^2 + m^2]^{\alpha}[(\mathbf{l}+\mathbf{q})^2 + m^2]^{\beta}}\right) = 0,$$

$$= -\frac{i}{2E\sqrt{1-\mathbf{u}^2}} \int \frac{\mathrm{d}^2\mathbf{l}}{(2\pi)^2} \frac{1}{[\mathbf{l}^2 + m^2][(\mathbf{l}+\mathbf{q})^2 + m^2]},$$

$$\int \frac{\mathrm{d}\omega}{2\pi} \frac{\omega^n}{[\omega - \mathbf{u}\cdot\mathbf{l} + i0][\omega + \mathbf{u}\cdot\mathbf{l} - i0]} = \frac{i(-1)^{n-1}}{2}\left(\mathbf{u}\cdot\mathbf{l} - i0\right)^{n-1}.$$

$$\int \frac{\mathrm{d}^2\mathbf{q}}{(2\pi)^2} e^{i\mathbf{q}\cdot\mathbf{x}} \int \frac{\mathrm{d}^2\mathbf{l}}{(2\pi)^2} \frac{m^2}{[\mathbf{l}^2 + m^2]^2[(\mathbf{l}+\mathbf{q})^2 + m^2]} = \frac{mr}{8\pi^2} K_0(mr) K_1(mr).$$

$$\int_0^{mr} \mathrm{d}r'\; r'\; K_0\left(mr'\right)^2 I_0\left(mr'\right) = \frac{\pi}{3\sqrt{3}m^2} - \int_{mr}^{\infty} \mathrm{d}r'\; r'\; K_0\left(mr'\right)^2 I_0\left(mr'\right),$$

$$a(\xi) = -Z_N \xi K_1(\xi) + \mathcal{O}\left(e^{-2\xi}\right),$$

$$\langle P(x) P^{\dagger}(x+R)\rangle_{N_t} = \sigma_0^{\frac{4-D}{2}} \frac{N_t}{\pi} \sum_{n=0}^{\infty} w_n \left(\frac{E_n}{2R}\right)^{\frac{1}{2}(D-3)} K_{\frac{1}{2}(D-3)}(E_n R)$$

$$E_n = \sigma_0 N_t \sqrt{1 + \frac{8\pi}{\sigma_0 N_t^2}\left[-\frac{1}{24}\left(D-2\right) + n\right]}.$$

$$\langle P(x)P^\dagger(x+R)\rangle_{N_t} \sim \sigma_0^{\frac{4-D}{2}} N_t \left(\frac{E_0}{R}\right)^{\frac{1}{2}(D-3)} K_{\frac{1}{2}(D-3)}(E_0 R)$$

$$\langle P(x)P^\dagger(x+R)\rangle_{N_t} \sim N_t\sqrt{\sigma_0}K_0(E_0 R)$$

$$\frac{T_{c,NG}}{\sqrt{\sigma_0}} = \sqrt{\frac{3}{\pi(D-2)}}$$

$$\Delta(g) = \frac{1}{\sqrt{(g)}}\partial_a[\sqrt{(g)}g^{ab}\partial_b]$$

$$\sigma(\beta) = \sigma_c t^{2\nu} \times (1 + at^\theta + bt)\ ,$$

$$\tilde{\gamma}_n{=}\gamma_n + (-1)^{(n+1)/2}\frac{1}{n2^{3n-1}}$$

$$\ddot{q} = -V'(q)$$

$$E = \frac{1}{2}\dot{q}^2 + V(q)$$

$$\partial_t^2\phi(t,x) - \partial_x^2\phi(t,x) = -V'(\phi)$$

$$i\hbar\frac{\partial\psi(t,\vec{x})}{\partial t} = \hat{H}\psi(t,\vec{x}).$$

$$\partial_t\psi(t\to t_c,\vec{x}_P)\to\infty.$$

$$\psi(t\to t_c,\vec{x}_P)\to\infty \quad \forall\vec{x}_P$$

$$\psi(t>t_0) = e^{i\hat{H}(t-t_0)}\psi(t_0)$$

$$\psi(t,\vec{x}) = \int d^3x' K(\vec{x},t;\vec{x'},t_0)\psi(t_0,\vec{x'})$$

$$\psi(t,\vec{x}) = \int d^3x' K(\vec{x},t;\vec{x'},t_c)\psi(t_c,\vec{x'})$$

$$t(\tau_b) = t(\tau_d).$$

$$\tau_c - \tau_b = N\epsilon, \quad \tau_d - \tau_c = N\epsilon, \quad \epsilon\to 0.$$

$$K_{II,III}(\tau_d,q_d;\tau_b,q_b) = \delta(q_b - q_d), \quad \tau_b\neq\tau_d.$$

$$S = \frac{1}{2}\int d^4x\left[\partial_\mu\phi\partial^\mu\phi - m^2\phi^2\right]$$

$$\ddot{q}(t) + \omega^2 q(t) = 0.$$

$$K(t_d,q_d;t_b,q_b) = \delta(q_d - q_b) \quad \text{at} \quad t_b = t_d$$

$$\tau_j = \frac{n_j^+ - n_j^-}{n_j}$$

$$\langle \tau_j \rangle = T_j, \qquad \mathrm{Var}(\tau_j) = \sigma_j^2 / n_j,$$

$$\langle \tau_j^2 \rangle = T_j^2 + \mathrm{Var}(\tau_j), \qquad \mathrm{Var}(\tau_j^2) = \frac{2(n_j - 1)(1 - T_j^2)[(2n_j - 3)T_j^2 + 1]}{n_j^3}.$$

$$\langle S \rangle = \frac{(n-1)S_M^\infty + M}{n}, \quad \text{with} \quad \mathrm{Var}(S) = \sum_{i=1}^{M} \mathrm{Var}(\tau_i^2),$$

$$\rho = p|\psi\rangle\langle\psi| + (1-p)\rho_{\mathrm{n}},$$

$$\mathrm{W} = \sigma_y \otimes \sigma_y \otimes \sigma_x - \sigma_x \otimes \sigma_x \otimes \sigma_x + 1,$$

$$\mathrm{p}_i(n_+) = \binom{n}{n_+} \left(\frac{1+T_i}{2}\right)^{n_+} \left(\frac{1-T_i}{2}\right)^{n_-},$$

$$\mathrm{P}(\tau_i) = \mathrm{p}_i\left(\frac{n}{2}(1+T_i)\right).$$

$$\mathrm{P}(\tau_i^2) = \mathrm{P}(\tau_i) + \mathrm{P}(-\tau_i) \quad \text{for} \quad \tau_i \neq 0,$$

$$\mathrm{P}(S) = \sum_{\tau_1^2 + \cdots + \tau_M^2 = S} \mathrm{P}(\tau_1^2) \cdots \mathrm{P}(\tau_M^2).$$

$$\mathrm{P}(Q^\infty \in \mathcal{O}_{\mathrm{acc}} | \mathrm{sep}) = 0.$$

$$\sum_{Q \in \mathcal{O}_{\mathrm{rej}}} \mathrm{P}(Q|\mathrm{sep}) \geq q_{\mathrm{F}}.$$

$$r = 1 - e_2 = \sum_{Q \in \mathcal{O}_{\mathrm{acc}}} \mathrm{P}(Q|\mathrm{ent}).$$

$$\mathrm{P}(\mathrm{ent}|Q) \geq q_{\mathrm{B}}, \quad \text{for all} \quad Q \in \mathcal{O}_{\mathrm{acc}},$$

$$L_Q = \begin{cases} \beta\, \mathrm{P}(\mathrm{sep}|Q) & \text{if } Q \in \mathcal{O}_{\mathrm{acc}} \\ (1-\beta)\, \mathrm{P}(\mathrm{ent}|Q) & \text{if } Q \in \mathcal{O}_{\mathrm{rej}}, \end{cases}$$

$$T_{\mu_1 \ldots \mu_N} = \langle R_{\mu_1} \ldots R_{\mu_N} \rangle = \mathrm{Tr}(\rho\, \sigma_{\mu_1} \otimes \cdots \otimes \sigma_{\mu_N}),$$

$$\langle E_{\mathrm{W}} \rangle = E_{\mathrm{W}}^\infty, \quad \text{and} \quad \mathrm{Var}(E_{\mathrm{W}}) = \sum_j \alpha_j^2 \mathrm{Var}(\tau_j).$$

$$\mathrm{P}(p) = \begin{cases} \mathcal{N} \mathrm{e}^{-\frac{(p-\bar{p})^2}{2(\Delta p)^2}} & p \in [p_{\min}, 1] \\ 0 & \text{else}, \end{cases}$$

$$\mathrm{W} = \sum_j \alpha_j \sigma_j.$$

$$E_{\mathrm{W}} = \mathrm{Tr}(\mathrm{W}\rho) = \sum_j \alpha_j T_j.$$

$$S \equiv \sum_{j=1}^{3^N} T_j^2,$$

$$\mathrm{P}(Q|\mathrm{sep}) = \sum_j \mathrm{P}(Q|\rho_j)\mathrm{P}(\rho_j|\mathrm{sep}),$$
$$\mathrm{P}(Q|\mathrm{ent}) = \sum_j \mathrm{P}(Q|\rho_j)\mathrm{P}(\rho_j|\mathrm{ent}).$$

$$\sum_{Q\in\mathcal{O}_{\mathrm{acc}}} \mathrm{P}(Q|\mathrm{sep}) \leq \sum_{Q\in\mathcal{O}_{\mathrm{acc}}} \mathrm{G}_{\mathrm{acc}}(Q|\mathrm{sep}) \leq 1 - q_{\mathrm{F}}.$$

$$\delta\Phi \leq \frac{1}{2\sqrt{\frac{t}{T_r}}\left(\frac{\partial P_{|1\rangle}}{\partial\Phi}\right)} = \frac{A}{\sqrt{t}},$$

$$P_{|1\rangle}(\Phi,\tau) = \frac{1}{2} + \frac{1}{2}e^{-A\tau - B^2\tau^2}\cos(\Delta\omega(\Phi)\tau),$$

$$P_{|1\rangle} = \frac{1}{2} + \frac{1}{2}\frac{e^{\frac{hf_q}{k_BT}} - 1}{e^{\frac{hf_q}{k_BT}} + 1}e^{-A\tau - B^2\tau^2}\cos(\Delta\omega\tau),$$

$$P_{|10..0\rangle} = \frac{1}{2} + \frac{1}{2}\frac{e^{\frac{hf_q}{k_BT}} - 1}{e^{\frac{hf_q}{k_BT}} + 1}e^{-N(A\tau + B^2\tau^2)}\cos(N\Delta\omega\tau).$$

$$\overline{\tau_l} = \frac{1}{F}\sum_{j=1}^{F}\frac{1}{M}\sum_{k=1}^{M}\tau_{j,k,l}, \quad \tau_{j,k,l} = \sum_{i=1}^{l}\tau_i^{(j,k)}n_i^{(j,k)}.$$

$$\overline{\left(\frac{\delta\Phi}{\Phi_0}\right)_l} = \sqrt{\frac{1}{\Phi_0^2 F}\sum_{j=1}^{F}\frac{1}{M-1}\sum_{k=1}^{M}(\hat{\Phi}_{jkl} - \Phi_j)^2}.$$

$$\vec{B}(x,y) = \frac{\mu_0}{4\pi}\int_l \frac{Id\vec{l}\times\vec{r}'}{|\vec{r}'|^3}$$

$$\Phi = \int_{x_1}^{x_2}\int_{y_1}^{y_2} B(x,y)dxdy.$$

$$S_x[\omega] = \frac{2k_BT\gamma}{M[(\omega_m^2 - \omega^2)^2 + \gamma^2\omega^2]}\,.$$

$$S_x[\omega] = \frac{2k_B T\gamma}{M[(\omega_m^2-\omega^2)^2+\gamma^2(1+g)^2\omega^2]}\,.$$

$$y_{\rm out} = \sqrt{\kappa} y_c - y_{\rm c,in}$$

$$A[\omega] = A_0 \exp\left(-i\phi\frac{\omega}{\omega_m}\right).$$

$$\chi_{\rm fb}[\omega] = \frac{1}{-i\omega\gamma_{\rm eff}+\omega_{\rm eff}^2-\omega^2}\,.$$

$$\phi_m = \tan^{-1}\left(-\frac{\kappa}{2\omega_m}\right)+\pi$$

$$\gamma_{\rm fb} = \frac{4GA_0}{\sqrt{\kappa^2+4\omega_m^2}}\,.$$

$$n_T = \frac{\gamma}{\gamma_{\rm eff}}\left(n_m^T+\frac{1}{2}\right).$$

$$n_{\rm qba} = C_{\rm eff}\frac{\kappa^2}{\kappa^2+4\omega_m^2}\,.$$

$$C_{\rm eff} = \frac{4G^2}{\kappa\gamma_{\rm eff}}\,.$$

$$n_m+\frac{1}{2} = n_T + n_{\rm qba} + n_{\rm fb}\,,$$

$$n_{m,\rm min}+\frac{1}{2} = \frac{1}{2}\sqrt{1+2n_{\rm add}}\,.$$

$$S_{\rm out}[\omega] = \langle a_{\rm out}^\dagger[\omega]a_{\rm out}[-\omega]\rangle+\frac{1}{2}\,.$$

$$\gamma_{\rm opt} = \frac{4G^2}{\kappa}\frac{1}{1+\left(\frac{\kappa}{4\omega_m}\right)^2}\,.$$

$$\gamma_{\rm eff} = \gamma-\gamma_{\rm opt}+\frac{4GA_0\left[(\kappa^2+8\omega_m^2)\sin\phi-2\kappa\omega_m\cos\phi\right]}{\kappa^3+16\kappa\omega_m^2}\,.$$

$$X_f = \chi_{\rm fb}\,,$$
$$X_{\rm inj} = -\frac{4G\omega_m\sqrt{\kappa}}{\kappa-2i\omega}\chi_{\rm fb}\,,$$

$$x[\omega] \equiv \frac{b[\omega]+b^\dagger[\omega]}{\sqrt{2}}\,,$$

$$x_\varphi[\omega] \equiv \frac{b[\omega]e^{i\varphi}+b^\dagger[\omega]e^{-i\varphi}}{\sqrt{2}}$$

$$s[\omega] = \frac{2\mathcal{G}\sqrt{\kappa}G}{\sqrt{\kappa^2/4+\omega_m^2}}\, x_{\phi_0+\phi_\tau}[\omega]\,.$$

$$A_0 = \frac{g_0\sqrt{\kappa}\alpha_0^2\mathcal{G}}{\sqrt{\kappa^2/4+\Delta_f^2}}D$$

$$\gamma_{\rm opt} = \mathcal{P}P\,,$$

$$G_{\rm rsb} = \frac{1}{2}\sqrt{\mathcal{P}P\kappa\left[1+\left(\frac{\kappa}{4\omega_m}\right)^2\right]}\,.$$

$$G = \frac{|\chi_c(\Delta)|}{|\chi_c(-\omega_m)|}G_{\rm rsb}\,,$$

$$\gamma_{\rm fb} = \mathcal{L}g\,.$$

$$A_0 = \frac{\mathcal{L}\mathcal{G}\sqrt{\kappa^2+4\omega_m^2}}{4G}\,.$$

$$\gamma_{\rm eff} = \gamma - \gamma_{\rm opt} + \gamma_{\rm fb,bsb}\,,$$

$$\gamma_{\rm fb,bsb} = \frac{4A_0G\sqrt{\kappa^4+20\kappa^2\omega_m^2+64\omega_m^4}}{\kappa^3+16\kappa\omega_m^2}\,.$$

$$-\sqrt{2}Ga^\dagger a = -\sqrt{2}G^2\kappa^2\mathcal{G}\mathcal{A}|\chi_c[\omega_m]\chi_c[\omega_f]|\left[|\chi_c[\omega_f-\omega_m]|x_{\Psi-}(t)\ +\ |\chi_c[\omega_f+\omega_m]|x_{\Psi+}(t)\right]$$

$$\Psi- = \phi_\tau+\phi_0-\phi_--\phi_1$$
$$\Psi_+ = \phi_\tau+\phi_0+\phi_++\phi_1$$

$$\hat{H}(x_1,x_2,\ldots x_N) = \sum_{i=1}^{N}\hat{h}(x_i) + \sum_{i<j=1}^{N}\hat{W}(x_i-x_j)$$

$$|\psi(t)\rangle = \sum_{\bar{n}} C_{\bar{n}}(t)|\bar{n};t\rangle,$$

$$|\bar{n};t\rangle = \prod_{i=1}^{M}\left(\frac{\left(b_i^\dagger(t)\right)^{n_i}}{\sqrt{n_i!}}\right)|vac\rangle.$$

$$\hat{W}(x_i-x_j) = \lambda\delta(x_i-x_j)$$

$$\rho^{(1)}(x_1'|x_1;t) = N\int dx_2\,dx_3...dx_N\,\psi^*(x_1',x_2,\ldots,x_N;t)$$
$$\psi(x_1,x_2,\ldots,x_N;t).$$

$$\rho^{(1)}(x_1'|x_1;t) = \sum_{kq} \rho_{kq}\phi_k^*(x_1';t)\phi_q(x_1;t)$$

$$\rho(x;t) = \rho^{(1)}(x_1' = x|x_1 = x;t)$$

$$\rho^{(p)}(x_1,\ldots,x_p;t) = \langle\psi(t)|\hat{\psi}^\dagger(x_1)\ldots\hat{\psi}^\dagger(x_p)\hat{\psi}(x_p) \\ \ldots\hat{\psi}(x_1)|\psi(t)\rangle$$

$$g^{(p)}(x_1,\ldots,x_p;t) = \frac{\rho^{(p)}(x_1,\ldots,x_p;t)}{\Pi_{i=1}^p|\rho^{(1)}(x_i)|}$$

$$\Lambda(t) = 1 - n_1(t)$$

$$H = \sum_{\langle xy\rangle}\left[-\frac{t}{2}(c_x^\dagger c_y + c_y^\dagger c_x) + V(n_x - \frac{1}{2})(n_y - \frac{1}{2})\right]$$

$$H_{N_f}^{(g)} = -\sum_{\langle xy\rangle}\prod_{f=1}^{N_f}\left(H_{\langle xy\rangle,f}^{(g)} + H_{\langle xy\rangle,f}^{(g),\mathrm{des}}\right)$$

$$H_{N_f=2}^{U(1)} \to H_{N_f=2}^{U(1)} + J\sum_{\langle xy\rangle} s_{xy,1}^3 s_{xy,2}^3,$$

$$\begin{aligned} w_A &= 1 \\ w_D &= \exp(\epsilon t)\sinh\epsilon t \\ w_A + w_D &= \exp(\epsilon t)\cosh\epsilon t, \end{aligned}$$

$$1 + \left(\frac{1}{2}e^{2\epsilon t} - \frac{1}{2}\right) = \left(\frac{1}{2}e^{2\epsilon t} + \frac{1}{2}\right)$$

$$\begin{aligned} &1 + \left(\frac{1}{2^{N_f}}e^{\epsilon(2t)^{N_f}} - \frac{1}{2^{N_f}}\right) \\ &= \left(\frac{1}{2^{N_f}}e^{\epsilon(2t)^{N_f}} + \left(1 - \frac{1}{2^{N_f}}\right)\right), \end{aligned}$$

$$\begin{aligned} H_e^s &= \sum_x s_{2x,2x+1,1}^3 s_{2x,2x+1,2}^3, \\ H_o^s &= \sum_x s_{2x+1,2x+2,1}^3 s_{2x+1,2x+2,2}^3. \end{aligned}$$

$$e^{\tau_1 J/4}e^{\tau_2 J/4}$$

$$e^{-\tau_1 J/4}e^{-\tau_2 J/4}$$

$$V \propto X_0 + X_0^\dagger.$$

$$\mathcal{R}^{(3)} \propto \langle[[[V(t_1+t_2+t_3), V(t_1+t_2)], V(t_1)], V]\rangle,$$

$$R_2 = \langle VV\left(t_1+t_2\right)V\left(t_1+t_2+t_3\right)V\left(t_1\right)\rangle\,,$$

$$R_3 = \langle VV\left(t_1\right)V\left(t_1+t_2+t_3\right)V\left(t_1+t_2\right)\rangle\,,$$

$$R_1^* = -\langle VV\left(t_1+t_2+t_3\right)V\left(t_1+t_2\right)V\left(t_1\right)\rangle\,,$$

$$R_1^* = -\langle \mathrm{FS}|\, X_0 e^{i\mathcal{H}_1(t_1+t_2+t_3)} X_0 e^{-i\mathcal{H}_2 t_3} X_0^\dagger e^{-i\mathcal{H}_1 t_2} X_0^\dagger\, |\mathrm{FS}\rangle\,,$$

$$R_1^* = -\sum_{nm\eta} \phi_0^{(n)} \Phi^{(n,\eta)} \Phi^{(m,\eta)*} \phi_0^{(m)*} e^{i\mathcal{E}_n t_1} e^{i[\mathcal{E}_n-\mathcal{E}_m]t_2} e^{i[\mathcal{E}_n-\mathcal{E}_\eta]t_3},$$

$$\int\int d\omega_1 d\omega_3 \mathcal{S}_{\mathrm{ESA}} = -\int\int d\omega_1 d\omega_3 \mathcal{S}_{\mathrm{ESE}} - \int\int d\omega_1 d\omega_3 \mathcal{S}_{\mathrm{GSB}}.$$

$$|n\rangle = \phi_0^{(n)}\,|0\rangle_1 + \sum_{k_p k_h} \phi_{k_p k_h}^{(n)}\,|k_p k_h\rangle_1\,,$$

$$|\eta\rangle = \sum_k{}' \Phi_k^{(\eta)}\,|k\rangle_2 + \sum_k{}' \sum_{k_p k_h} \Phi_{k k_p k_h}^{(\eta)}\,|k k_p k_h\rangle_2\,,$$

$$|k k_p k_h\rangle_2 = C_{k,k_p,k_h} X_{-k}^\dagger X_{k-k_p+k_h}^\dagger c_{k_p}^\dagger c_{k_h}\,|\mathrm{FS}\rangle$$

$$\boldsymbol{r} = a(t)[\boldsymbol{q} - D_+(t)\boldsymbol{\nabla}\Phi],$$

$$\Phi(\boldsymbol{q}) = \Phi(\boldsymbol{q}_{cm}) + \Delta q^i \left.\frac{\partial\Phi}{\partial q^i}\right|_{q_{cm}} + \Delta q^i \Delta q^j \left.\frac{\partial^2\Phi}{\partial q_i \partial q_j}\right|_{q_{cm}}$$

$$L_i \approx -t_m \epsilon_{ijk} I_{jl} T_{lk}$$

$$\tau_i = \epsilon_{ijk} I_{jl} \partial_l \partial_k \Phi.$$

$$L_i = \epsilon_{ijk} I_{jl} \mathcal{M}_{lk}$$

$$\mathcal{M}_{ij} = \frac{2GM_2}{b^2 v}(\hat{b}_i \hat{b}_j - \hat{c}_i \hat{c}_j)$$

$$\lambda = \frac{L\sqrt{|E|}}{GM^{5/2}},$$

$$\lambda_{TLT} \sim \left(\frac{GM_2}{v^2 b}\right)^{1/2} \left(\frac{R_1}{b}\right)^{3/2}$$

$$i\frac{d}{dr}\begin{bmatrix} \nu_L \\ \nu_R \end{bmatrix} = \begin{bmatrix} H_L + (\dot{\phi}/2)I & \mu B(r) \\ \mu^\dagger B(r) & H_R - (\dot{\phi}/2)I \end{bmatrix} \begin{bmatrix} \nu_L \\ \nu_R \end{bmatrix},$$

$$V_\alpha + \dot{\phi} = 0.$$

$$\gamma_\alpha = \frac{2(2\mu_\nu B)^2}{|\dot{V}_\alpha + \ddot{\phi}|}.$$

$$\Phi_e^{NO} = |U_{e3}|^2\Phi_{ie} + (1-|U_{e3}|^2)\Phi_{ix} \approx 0.02\Phi_{ie} + 0.98\Phi_{ix},$$

$$\Phi_e^{IO} = |U_{e2}|^2\Phi_{ie} + (1-|U_{e2}|^2)\Phi_{ix} \approx 0.3\Phi_{ie} + 0.7\Phi_{ix}.$$

$$\Phi_{\bar{e}}^{NO} = |U_{e1}|^2\Phi_{i\bar{e}} + (1-|U_{e1}|^2)\Phi_{ix} \approx 0.7\Phi_{i\bar{e}} + 0.3\Phi_{ix},$$

$$\Phi_{\bar{e}}^{IO} = |U_{e3}|^2\Phi_{i\bar{e}} + (1-|U_{e3}|^2)\Phi_{ix} \approx 0.02\Phi_{i\bar{e}} + 0.98\Phi_{ix}.$$

$$\Phi_{i\alpha} \to e^{-\frac{\pi}{2}\gamma_\alpha}\Phi_{i\alpha},$$

$$\frac{dN_{\nu_\alpha}}{dE_r} = \frac{N_{\rm t}}{4\pi R^2}\int dE_t \Phi_\alpha\left(E_t\right)\sigma_\alpha\left(E_t\right)W\left(E_r,E_t\right)$$

$$\chi^2 = \min_{\xi}\sum_{i=1}^{n} 2\left[(1+\xi)F_i - D_i + D_i\ln\left(\frac{D_i}{(1+\xi)F_i}\right)\right],$$

$$\tilde{\phi} \;\equiv\; A(\chi)\phi\,, \qquad \tilde{\psi} \;\equiv\; A^{3/2}(\chi)\psi\,,$$

$$A^2(\chi) \;=\; 1 \;+\; b\,\frac{\chi}{M} \;+\; c\,\frac{\chi^2}{M^2} \;+\; \mathcal{O}\!\left(\frac{\chi^3}{M^3}\right),$$

$$V(\chi) \;=\; \frac{1}{2}\,\mu_\chi^2\,\chi^2\,.$$

$$\tilde{\chi} \;\equiv\; \left(1 \;+\; \frac{b^2\tilde{\phi}^2}{4M^2}\right)^{1/2}\chi\,,$$

$$V(\tilde{\chi}) = \frac{1}{2}\mu_\chi^2\tilde{\chi}^2 + \frac{3}{2}\,\frac{\mu^4}{\lambda}\left(1 + 2b\,\frac{\tilde{\chi}}{M} + 2c\,\frac{\tilde{\chi}^2}{M^2} + b^2\,\frac{\tilde{\chi}^2}{M^2}\right)\,,$$

$$\frac{\mu^2\tilde{\phi}_m^4}{v^2} - \tilde{\phi}_m^2\left[\mu^2\left(1 + \frac{b\tilde{\chi}_m}{M} + \frac{c\tilde{\chi}_m^2}{M^2}\right) + \frac{b\mu_\chi^2\tilde{\chi}_m^2}{4M^2}\right] + \rho_\psi = 0\,,$$

$$\tilde{\chi}_m\left[\frac{\mu^2 v^2}{M^2}\left(c - \frac{c\tilde{\phi}_m^2}{v^2} + \frac{b^2}{2}\right) + \mu_\chi^2\left(1 - \frac{b^2\tilde{\phi}_m^2}{4M^2}\right)\right] = \frac{b\mu^2 v^2}{2M}\left(1 - \frac{\tilde{\phi}_m^2}{v^2}\right)\,,$$

$$\frac{\tilde{\chi}_m}{M} = -\frac{b\rho_\psi}{2\mu_\chi^2 M^2 + (2c-b^2)\rho_\psi}\,.$$

$$\theta \approx -\frac{v}{2M}\left[b + \frac{(4c-b^2)\tilde{\chi}_m}{2M} + \frac{\rho_\psi}{\mu^2 v^2}\left(b + \frac{2c\tilde{\chi}_m}{M} - \frac{5b^2\tilde{\chi}_m}{2M}\right)\right]\,.$$

$$\beta(\rho_\psi) = \frac{2b\mu_\chi^2 M^2}{2\mu_\chi^2 M^2 + (2c-b^2)\rho_\psi} + \mathcal{O}\left(\frac{\rho_\psi}{\mu^2 v^2}\right)\,.$$

$$F_s = -\frac{\beta(\rho_\psi)}{M}\nabla s \ .$$

$$\frac{\rho_\psi}{\mathrm{gcm}^{-3}} \gtrsim \frac{0.46}{2c-b^2}\left(\frac{\mu_\chi}{\mathrm{eV}}\right)^2\left(\frac{M}{\mathrm{GeV}}\right)^2 \ ,$$

$$\beta(\rho_\psi) \approx \frac{2b\mu_\chi^2 M^2}{(2c-b^2)\rho_\psi} \ ,$$

$$\frac{1}{(2c-b^2)^{1/2}}\left(\frac{\mu_\chi}{\mathrm{eV}}\right)\left(\frac{M}{\mathrm{GeV}}\right) \lesssim 2.1\times 10^{-13} \ .$$

$$\frac{1}{(2c-b^2)^{1/2}}\left(\frac{\mu_\chi}{\mathrm{eV}}\right)\left(\frac{M}{\mathrm{GeV}}\right) < 5.1 \ .$$

$$\frac{M_c}{R} \gg \frac{|b^2-2c|M^2}{b^2} \ .$$

$$2.1\times 10^{-13} \lesssim \frac{1}{(2c-b)^{1/2}}\left(\frac{\mu_\chi}{\mathrm{eV}}\right)\left(\frac{M}{\mathrm{GeV}}\right) \lesssim 5.1 \ .$$

$$\frac{M}{\mathrm{GeV}} \ll \frac{(3\times 10^{14})b}{|b^2-2c|^{1/2}} \ .$$

$$\alpha_{hs} = \frac{b\mu^2}{2M} \ ,$$
$$\lambda_{hs} = \frac{c\mu^2}{2M^2} + \frac{b^2\mu_\chi^2}{8M^2} \ .$$

$$h = \delta\tilde{\phi}\cos\theta + \delta\tilde{\chi}\sin\theta \ ,$$
$$s = \delta\tilde{\chi}\cos\theta - \delta\tilde{\phi}\sin\theta \ .$$

$$d(y,\vec{C}) - d(y,\vec{Y}) \leq cost(C_j,\vec{C}) - cost(C_j,\vec{Y})$$

$$d(C_i,C_{i+1}) > \Big(\frac{\gamma^2+1}{\gamma+1}\frac{1}{c} - 1\Big)D(C_i)$$

$$d(C_i,C_{i+1}) > \Big(\frac{c-1}{c}(\gamma-1)\Big)D(C_i)$$

$$d(C_i,C_{i+1}) > \max\Big\{\frac{\gamma^2+1}{\gamma+1}\frac{1}{c} - 1, \frac{c-1}{c}(\gamma-1)\Big\}D(C_i)$$

$$c = \frac{2\gamma^2}{\gamma^2+\gamma}$$

$$w > \frac{\gamma^2-\gamma}{\gamma+1}\frac{y}{2} + (\gamma-2)\frac{z}{2}$$

$$D(C_i) = d(x_{i,l}, x_{i,r}) = d(c_i, x_{i,r}) < \frac{1}{(\gamma-1)} d(x_{i,r}, x_{i+1,r}) = \frac{1}{(\gamma-1)} d(C_i, C_{i+1}) \Rightarrow$$
$$d(C_i, C_{i+1}) > (\gamma-1) D(C_i)$$

$$\frac{\gamma^2+1}{\gamma+1}\frac{1}{c} - 1 = \frac{c-1}{c}(\gamma-1)$$

$$d(C_i, C_{i+1}) > \Big(\frac{\gamma^2+1}{2\gamma} - 1\Big) D(C_i) = \frac{(\gamma-1)^2}{2\gamma} D(C_i)$$

$$cost(\vec{x}, M(\vec{x})) = \sum_{i=1}^{n} \mathbb{E}_{\vec{c} \sim M(\vec{x})}(d(x_i, \vec{c}))\,.$$

$$\gamma - 1 \geq \frac{\gamma^2+1}{2\gamma} - 1$$

$$D(C_i) = d(x_{i,l}, x_{i,r}) = d(c_i, x_{i,r}) < \frac{1}{(\gamma-1)} d(x_{i,r}, x_{i-1,r}) = \frac{1}{(\gamma-1)} d(C_i, C_{i+1}) < d(C_i, C_{i+1})$$

$$D(C_i) + d(C_i, C_{i+1}) = d(x_{i,l}, c_i) + d(c_i, x_{i+1,l}) = d(x_{i,l}, x_{i+1,l}) > (\gamma-1) d(x_{i,l}, c_i) \geq 2D(C_i)\,,$$

$$\frac{\gamma^2-3\gamma-2}{\gamma+1}\frac{(\gamma-2)}{2} + \frac{\gamma-2}{2} \geq 1 \Rightarrow$$
$$\gamma^3 - 4\gamma^2 + \gamma \geq 0$$

$$\tilde{Q} := \begin{bmatrix} Q & \mathbf{u} \\ \mathbf{v}^T & \tau \end{bmatrix}.$$

$$\text{T-value}(\mathbf{x}_i) = \frac{\mu_{i,\diamond} - \mu_{i,\oslash}}{\sqrt{(\sigma_{i,\diamond}^2 + \sigma_{i,\oslash}^2)/M}}.$$

$$\text{MV}_\xi(\mathbf{x}) := \arg\max_{y \in \mathcal{Y}} \mathbb{P}_{\theta \sim \xi}\left[h_\theta(\mathbf{x}) = y\right].$$

$$\left|\hat{\theta} - \frac{\Delta(\mu_+ - \mu_-)}{2} - \frac{(\mu_+ + \mu_-)}{2}\right| \leq \delta$$

$$P(E) \geq 1 - 2\exp\Big\{-\frac{2\delta^2}{9\sigma^2[1/\tilde{n}_+ + 1/\tilde{n}_-]}\Big\} - \frac{9\texttt{Var}(I_i^+)}{4\delta^2} - \frac{9\texttt{Var}(I_i^-)}{4\delta^2}.$$

$$\text{minimize}_{Q \in \mathbb{R}^{N \times K}} \langle Q, C \rangle$$

$$\mathcal{L}(Q) = \text{minimize}_{Q \in \mathbb{R}^{N \times K}} \langle Q, C \rangle - \epsilon \mathbb{H}(Q)$$

$$\mathbf{a} = \exp\left(\frac{\lambda^{(1)}}{\epsilon} - 1/2\right)$$

$$\mathbf{b} = \exp\left(\frac{\lambda^{(2)}}{\epsilon} - 1/2\right).$$

$$Q^* = \operatorname{diag}(\mathbf{a}^{(J)}) \exp\left(-\frac{C_{i,k}}{\epsilon}\right) \operatorname{diag}(\mathbf{b}^{(J)}).$$

$$\Delta_{\mathcal{D}}(\theta) \leq \sqrt{\frac{\gamma_{\mathcal{F}}(\xi, \pi) + \ln(N/\delta)}{2(N-1)}}$$

$$\mu_{i,\diamond} = \frac{1}{M} \sum_{m=1}^{M} p_m(y = \diamond \mid \mathbf{x}_i)$$

$$\mu_{i,\oslash} = \frac{1}{M} \sum_{m=1}^{M} p_m(y = \oslash \mid \mathbf{x}_i)$$

$$\sigma^2_{i,\diamond} = \frac{1}{M} \sum_{m=1}^{M} \Big(p_m(y = \diamond \mid \mathbf{x}_i) - \mu_{i,\diamond}\Big)^2$$

$$\sigma^2_{i,\oslash} = \frac{1}{M} \sum_{m=1}^{M} \Big(p_m(y = \oslash \mid \mathbf{x}_i) - \mu_{i,\oslash}\Big)^2.$$

$$\sum_{k=1}^{K} \left|\hat{\theta}_k - \mu_k\right| \leq \delta$$

$$\sum_{k=1}^{K} \left|\hat{\theta}_k - \mu_k\right| = \sum_{k=1}^{K} \left|\hat{\theta}_k - \mu_k\right| - \sum_{k=1}^{K} \mathbb{E}(I^k) \left[\mu_k - \mu_{\setminus k}\right].$$

$$P(E) \geq 1 - 2\sum_{k=1}^{K}\sum_{j=1}^{d} \exp\Big\{-\frac{\delta^2 \tilde{n}_k}{8\sigma_j^2}\Big\} - \sum_{k=1}^{K} \frac{4\mathtt{Var}(I^k)}{\delta^2} \left|\mu_k - \mu_{\setminus k}\right|.$$

$$\mathbb{H}\left[p(y \mid \mathbf{x}, D)\right] \sim \mathbb{H}\left[\frac{1}{M} \sum_{m=1}^{M} \underbrace{p_m\Big(y = 1, ..., K \mid \mathbf{x}\Big)}_{\in \mathbb{R}^K}\right]$$

$$L(\mathbf{x}; \tilde{q}) := \sum_{j=1}^{M} q_j P(y = k \mid \mathbf{x}, \theta_j),$$

$$\arg\max_{\mathbf{x} \in \mathcal{D}_u} L(\mathbf{x}; \tilde{q}).$$

$$\arg\max_{\mathbf{x} \in \mathcal{D}_u} \min_{\tilde{q} \in B(\varepsilon)} L(\mathbf{x}; \tilde{q}),$$

$$\max_{\tilde{q} \in B(\varepsilon)} L(\mathbf{x}; \tilde{q}) = \frac{1}{M} \sum_{m=1}^{M} p_{mk} - \sqrt{\varepsilon \sum_{m=1}^{M} \left(\frac{1}{M} p_{mk} - p_{mk}\right)^2}.$$

$$C := \begin{bmatrix} -\log \bar{P} & \mathbf{0}_N \\ \mathbf{0}_K^T & 0 \end{bmatrix}$$

$$L(\mathbf{x}_k, t_k) - L(\mathbf{x}_k, t_k - \Delta t_k) = p_k\, C,$$

$$e_k \doteq (\mathbf{x}_k, t_k, p_k) \quad \overset{\mathbf{W}}{\mapsto} \quad e'_k \doteq (\mathbf{x}'_k, t_{\mathrm{ref}}, p_k),$$

$$I(\mathbf{x}; \boldsymbol{\theta}) \doteq \sum_{k=1}^{N_e} b_k\, \delta(\mathbf{x} - \mathbf{x}'_k(\boldsymbol{\theta})).$$

$$G(\boldsymbol{\theta}) \equiv \mathrm{Var}\big(I(\mathbf{x}; \boldsymbol{\theta})\big) \doteq \frac{1}{|\Omega|} \int_\Omega (I(\mathbf{x}; \boldsymbol{\theta}) - \mu_I)^2 d\mathbf{x},$$

$$\mathbf{x}'_k = (1 - \tilde{t}_k h_z)\, \mathbf{x}_k,$$

$$\mathcal{D}(\mathcal{E}, \boldsymbol{\theta}) \doteq \{\nabla \cdot \mathbf{f}_k\}_{k=1}^{N_e},$$

$$\mathrm{DIWE}(\mathbf{x}; \mathcal{E}, \boldsymbol{\theta}) \doteq \frac{1}{N_e(\mathbf{x})} \sum_k (\nabla \cdot \mathbf{f}_k)\, \delta(\mathbf{x} - \mathbf{x}'_k),$$

$$R_{\mathrm{DIWE}}(\mathcal{E}, \boldsymbol{\theta}) \doteq \frac{1}{|\Omega|} \int_\Omega \mathrm{DIWE}(\mathbf{x}; \mathcal{E}, \boldsymbol{\theta}) d\mathbf{x}.$$

$$\mathcal{A}(\mathcal{E}, \boldsymbol{\theta}) \doteq \big\{|\det(\mathtt{J}(e_k))|\big\}_{k=1}^{N_e},$$

$$\mathrm{IWA}(\mathbf{x}) \doteq 1 + \frac{1}{N_e(\mathbf{x})} \sum_{k=1}^{N_e} \big(|\det(\mathtt{J}(e_k))| - 1\big)\, \delta(\mathbf{x} - \mathbf{x}'_k).$$

$$R_{\mathrm{IWA}}(\mathcal{E}, \boldsymbol{\theta}) \doteq \frac{1}{|\Omega|} \int_\Omega \mathrm{IWA}(\mathbf{x}; \mathcal{E}, \boldsymbol{\theta}) d\mathbf{x}.$$

$$\mathbf{x}^{h\prime} \sim \mathtt{R}(t\boldsymbol{\omega})\, \mathbf{x}^h,$$

$$\det(\mathtt{J}) = (\mathbf{r}_3^\top \mathbf{x}^h)^{-3}.$$

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\mathrm{argmin}}\, J(\boldsymbol{\theta}) = \underset{\boldsymbol{\theta}}{\mathrm{argmin}}\, (-G(\boldsymbol{\theta}) + \lambda R(\boldsymbol{\theta}))\,.$$

$$\begin{pmatrix} \mathbf{x}'_k \\ 1 \end{pmatrix} \sim \begin{pmatrix} \mathtt{R}(t_k \omega_Z) & t_k \mathbf{v} \\ \mathbf{0}^\top & 1 \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{x}_k \\ 1 \end{pmatrix},$$

$$\begin{pmatrix} \mathtt{A} & \mathbf{b} \\ \mathbf{0}^\top & 1 \end{pmatrix}^{-1} = \begin{pmatrix} \mathtt{A}^{-1} & -\mathtt{A}^{-1}\mathbf{b} \\ \mathbf{0}^\top & 1 \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{x}'_k \\ 1 \end{pmatrix} \sim \begin{pmatrix} \mathtt{R}(-t_k \omega_Z) & -\mathtt{R}(-t_k \omega_Z)(t_k \mathbf{v}) \\ \mathbf{0}^\top & 1 \end{pmatrix} \begin{pmatrix} \mathbf{x}_k \\ 1 \end{pmatrix}.$$

$$\mathbf{x}'_k = \mathtt{R}(-t_k \omega_Z)(\mathbf{x}_k - t_k \mathbf{v}).$$

$$\mathtt{J}_k = \frac{\partial \mathbf{x}'_k}{\partial \mathbf{x}_k} = \mathtt{R}(-t_k \omega_Z),$$

$$\nabla \cdot \mathbf{f} = -2\omega_Z \sin(t\omega_Z).$$

$$\mathbf{x}_k^{h\prime} \sim \mathtt{R}(t_k \boldsymbol{\omega})\, \mathbf{x}_k^h,$$

$$\det(\mathtt{J}_k) = (\mathbf{r}_{3,k}^\top \mathbf{x}_k^h)^{-3}.$$

$$\mathbf{x}_k' = \mathbf{x}_k - t_k \left(\mathbf{v} + (h_z+1)\mathtt{R}(\phi)\mathbf{x}_k - \mathbf{x}_k\right)$$

$$\det(\mathtt{J}_k) = (1+t_k)^2 - 2(1+t_k)t_k(h_z+1)\cos\phi + t_k^2(h_z+1)^2.$$

$$\begin{pmatrix} \mathbf{x}_k' \\ 1 \end{pmatrix} \sim \begin{pmatrix} \beta_k \mathtt{R}(t_k\omega_Z) & t_k\mathbf{v} \\ \mathbf{0}^\top & 1 \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{x}_k \\ 1 \end{pmatrix}.$$

$$\begin{pmatrix} \mathbf{x}_k' \\ 1 \end{pmatrix} \sim \begin{pmatrix} \beta_k^{-1} \mathtt{R}(-t_k\omega_Z) & -\beta_k^{-1}\mathtt{R}(-t_k\omega_Z)(t_k\mathbf{v}) \\ \mathbf{0}^\top & 1 \end{pmatrix} \begin{pmatrix} \mathbf{x}_k \\ 1 \end{pmatrix}.$$

$$\mathbf{x}_k' = \beta_k^{-1}\mathtt{R}(-t_k\omega_Z)(\mathbf{x}_k - t_k\mathbf{v}).$$

$$\det(\mathtt{J}_k) = \beta_k^{-2} = \frac{1}{(1+t_k s)^2}.$$

$$R_D(\mathcal{E}, \boldsymbol{\theta}) \doteq \frac{1}{N_e}\sum_{k=1}^{N_e} \nabla \cdot \mathbf{f}_k. \quad \text{(mean)}$$

$$R_A(\mathcal{E}, \boldsymbol{\theta}) \doteq \frac{1}{N_e}\sum_{k=1}^{N_e} |\det(\mathtt{J}(e_k))| \quad \text{(mean)}$$

$$S_k = \sum_i^{\mathrm{C}} \sum_j^{\mathrm{C}} |R_{k,i,j}|^2,$$

$$s_p = [S_1, S_2, ..., S_p],$$

$$g(K, s_p) = \frac{\sum_{k=1}^K S_k}{\sum_{k=1}^p S_k}.$$

$$\frac{\partial u}{\partial t} = -u\cdot\nabla u - \nabla p + \frac{1}{Re}\Delta u + \sin(ny)\hat{x}, \quad \nabla \cdot u = 0, \quad \text{on } [0,2\pi]^2 \times (0,\infty)$$

$$\begin{aligned} \partial_t u(x,t) + \partial_x\left(u^2(x,t)/2\right) &= \nu \partial_{xx} u(x,t), \quad t \in (0,1] \\ u(x,0) &= u_0(x), \qquad\quad x \in (0,1) \end{aligned}$$

$$\begin{aligned} -\nabla\cdot(a(x)\nabla u(x)) &= f(x), \quad x \in (0,1)^2 \\ u(x) &= 0, \qquad\quad x \in \partial(0,1)^2 \end{aligned}$$

$$\begin{aligned}\partial_t w(x,t) + u(x,t)\cdot\nabla w(x,t) &= \nu\Delta w(x,t) + f(x),\\ \nabla\cdot u(x,t) &= 0, \quad x\in(0,1)^2\\ w(x,0) &= w_0(x), \quad t\in(0,T]\end{aligned}$$

$$\mathcal{G}_\theta := \mathcal{Q}\circ(W_L+\mathcal{K}_L)\circ\cdots\circ\sigma\left(W_1+\mathcal{K}_1\right)\circ\mathcal{P},$$

$$\left(\mathcal{K}v_t\right)(x) = \mathcal{F}^{-1}\left(R\cdot T_K\left(\mathcal{F}v_t\right)\right)(x) \quad \forall x\in D,$$

$$FMf\circ\otimes_X^M = \otimes_Y^M\circ MFf.$$

$$\begin{aligned}&LFX\to MFX\to FMX\to FMY\\ &= LFX\to FLX\to FLY\to FMY\\ &= LFX\to LFY\to FLY\to FMY\\ &= LFX\to MFX\to MFY\to FMY\end{aligned}$$

$$(\phi'\circ\phi)(\psi'\circ\psi) = (\phi'\psi')\circ(\phi\psi) : FG\to F''G''.$$

$$\phi\psi : FH\to GK = \phi K\circ F\psi = G\psi\circ\phi H$$

$$\otimes_X^L(\bar{f})(\bar{a}) = \prod_{i=1}^{n} f_i(a_i)$$

$$MX = \{m\in FX \mid \forall a\in X\ m(a)\in\mathbb{N}\}.$$

$$DX = \{d\in FX \mid \sum_{a\in X} d(a) = 1\}.$$

$$X^{(n)} = \{m\in MX \mid \sum_{a\in X} m(a) = n\},$$

$$\begin{aligned}&LLF\to MMF\to MFM\to FMM\to FM\\ &= LLF\to MMF\to MF\to FM\end{aligned}$$

$$\otimes_Y^L(LFh(\bar{f}))(\bar{b}) = FLh(\otimes_X^L(\bar{f}))(\bar{b}).$$

$$\otimes_Y^L\circ LFh = FLh\circ\otimes_X^L.$$

$$F\mu_X^L\circ\otimes_{LX}^L\circ L\otimes_X^L = \otimes_X^L\circ\mu_{FX}^L.$$

$$\mu_{LX}^F\circ F\otimes_X^L\circ\otimes_{FX}^L = \otimes_X^L\circ L\mu_X^F.$$

$$\sum_{i:R=1} w_i\phi(X_i) = \frac{1}{N_{pop}}\sum_{j=1}^{N_{pop}}\phi(X_j)$$

$$k(X_i,X_j) = exp(-||X_i-X_j||^2/b)$$

$$\min_{w} \; D(w,q) \quad \text{s.t.} \sum_{i:R_i=1} w_i \phi(X_i) = T, \; \sum_{i:R_i=1} w_i = 1, \text{ and } \; 0 \leq w_i \leq 1$$

$$\sum_{i:R=1} w_i K_i = \frac{1}{N_{pop}} \sum_{j=1}^{N_{pop}} K_j, \; s.t. \sum_i w_i = 1, \; w_i \geq 0, \; \forall i$$

$$\sum_{i:R=1} w_i K_i = \sum_{j=1}^{N_{pop}} w_j^{(pop)} K_j, \; s.t. \sum_i w_i = 1, \; w_i \geq 0, \; \forall i$$

$$\sqrt{\gamma} || (w^{(pop)\top} \mathbf{V}_{pop} - w_s^\top \mathbf{V}_s) \mathbf{A}^{1/2} ||_2$$

$$\sum_{x \in X} \phi(x) w(x) p(X = x \mid R = 1) = \sum_{x \in X} \phi(x) p(X = x)$$

$$w(x) = \frac{p(x)}{p(x \mid R = 1)} = \frac{p(x)p(R=1)}{p(R=1|x)p(x)} = \frac{p(R=1)}{p(R=1|x)}$$

$$w_i = \frac{N_s / N_{pop}}{g(\phi(X_i)^\top \theta + \eta_i)}$$

$$w(x) = \frac{p(x \mid R = 0)}{p(x \mid R = 1)} = \frac{p(R=0|x)p(R=1)}{p(R=1|x)p(R=0)} = \frac{p(R=0|x)}{p(R=1|x)} \frac{N_s}{N_{pop}}$$

$$\mu_s(w) = \frac{1}{N_s} \sum Y_i = \frac{1}{N_s} \sum w_i \phi(\mathbf{X})_i^\top \beta + \frac{1}{N_s} \sum w_i Z_i + \frac{1}{N_s} \sum w_i \nu_i$$

$$\mu = \overline{\phi(X)}_{pop}^\top \beta + \frac{1}{N_{pop}} \sum Z_i + \nu_i$$

$$\sum_{x \in X} \phi(x) w(x) p(X = x \mid R = 1) = \sum_{x \in X} \phi(x) p(X = x \mid R = 0)$$

$$\begin{aligned} p(S = 1) = logit^{-1} \Big(& PID(3way) + Age(4way) + Gender + educ(3way) + Race(4way) \\ & + BornAgain + PID(3way) * Age(4way) + BornAgain * Age(4way) \Big) \end{aligned}$$

$$\begin{aligned} p(Vote = D) = & PID(3way) + Age(4way) + Gender + educ(3way) + Race(4way) \\ & + BornAgain + PID(3way) * Age(4way) + BornAgain * Age(4way) \end{aligned}$$

$$\hat{\mu} = \sum_{i:R_i=1} w_i Y_i$$

$$Y = \phi(X)^\top \beta + Z + \nu$$

$$w_i = h(\phi(X_i)^\top \theta + \eta_i) = \frac{1 - g(\phi(X_i)^\top \theta + \eta_i)}{g(\phi(X_i)^\top \theta + \eta_i)} \frac{N_s}{N_{pop}} = e^{-(\phi(X_i)^\top \theta + \eta_i)} \frac{N_s}{N_{pop}}$$

$$Y_i = \sum_{j=1}^{p} \beta_j X_{ij} + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2),$$

$$\beta_j \sim (1-\gamma_j)N(0, \tau_j^2) + \gamma_j N(0, c_j^2 \tau_j^2);$$
$$\gamma_j \sim Bern(\pi_j),$$

$$Y_i = \sum_{j=1}^{p} \beta_j \gamma_j X_{ij} + \epsilon_i, \epsilon_i \sim N(0, \sigma^2);$$
$$\beta_j \sim N(0, s^2), \qquad \gamma_j \sim Bern(\pi_j).$$

$$\beta_{gj} = \gamma_g^{(1)} \gamma_{gj}^{(2)} b_{gj}, \quad \gamma_g^{(1)} \sim Bern(\pi_1), \quad \gamma_{gj}^{(2)} \sim Bern(\pi_2),$$
$$b_{gj} \sim N(0, s^2),$$

$$\mathbf{Y} = \mathbf{XB} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim MN_{n\times q}(\mathbf{0}, \mathbf{I}_n \otimes \Sigma),$$

$$\gamma_{gj} \sim \text{Bern}(\pi_{gj}^{(\gamma)}), \quad \Phi^{-1}(\pi_{gj}^{(\gamma)}) = d_0 + d_1 A_{gj}, \quad d_0 \sim N(0, \sigma^2),$$
$$d_1 \sim N(\mu_d, \tau^2),$$

$$\beta_{gj,k}|\mathbf{X}, \mathbf{Y} \sim l_{gj,k} N((1-D_n)\hat{\beta}_{gj,k}^{LS}, \frac{1-D_n}{n}\Sigma_{kk}) + (1-l_{gj,k})\delta_0(\beta_{gj,k}),$$

$$BFDR(t) = \frac{\sum_{g=1}^{G}\sum_{j=1}^{m_g} P_{gj}(H_0|\mathbf{X}, \mathbf{Y}) d_{gj}(t)}{\sum_{g=1}^{G}\sum_{j=1}^{m_g} d_{gj}(t)}.$$

$$s^2|rest \sim IG\left(1 + \frac{pq}{2}, \nu + \frac{1}{2}Tr(\mathbf{B}\Sigma^{-1}\mathbf{B}^T)\right)$$

$$\alpha_g|rest \sim \text{Bern}(\pi_*^{(\alpha)}),$$

$$\gamma_{gj}|rest \sim \text{Bern}(\pi_{g*}^{(\gamma)}),$$

$$\omega_{gj,k}|rest \sim \text{Bern}(\pi_{gj*}^{(\omega)}),$$

$$\Sigma|rest \sim \text{IW}\left(p+q+n, \mathbf{I} + (\mathbf{Y}-\mathbf{XB})^T(\mathbf{Y}-\mathbf{XB}) + s^2\mathbf{B}^T\mathbf{B}\right)$$

$$Z_{gj}^{\mathcal{M}_S} = \frac{PIP_{gj}^{\mathcal{M}_S} - \frac{1}{R}\sum_{r=1}^{R} PIP_{gj}^{\mathcal{M}_{S,r}}}{\sqrt{\frac{1}{R}\sum_{r=1}^{R}(PIP_{gj}^{\mathcal{M}_{S,r}} - \left(\frac{1}{R}\sum_{r=1}^{R} PIP_{gj}^{\mathcal{M}_{S,r}}\right))^2}},$$

$$(d_0, d_1)|rest \sim N\left((\Sigma_d^{-1} + \mathbf{A}^T\mathbf{A})^{-1}(\Sigma_d \vec{M}_d + \mathbf{A}^T\vec{\xi}), (\Sigma_d^{-1} + \mathbf{A}^T\mathbf{A})^{-1}\right)$$

$$\nu^{(m)} = \frac{1}{\text{E}_{\nu^{(m-1)}}\left[\frac{1}{s^2}|\mathbf{Y}\right]},$$

$$\vec{b}_{gj}|rest \sim N(\vec{M}_{gj}, \Sigma_{gj}),$$

$b_{gj,k}|\mathbf{X},\mathbf{Y} \sim N(M_{gj,k}, \Sigma_{gj,kk})$.

$$\lim_{n\to\infty} \frac{(1+ns^2)^{1/2}}{\exp\left(-\frac{1-D_n}{2\Sigma_{kk}}n\right)} = \lim_{n\to\infty} \frac{(s^2/2)(1+ns^2)^{-1/2}}{(-\frac{1-D_n}{2\Sigma_{kk}})\exp\left(-\frac{1-D_n}{2\Sigma_{kk}}n\right)} = 0,$$

$$m(u,v) = \frac{1}{p}\mathbb{E}\left[\langle \mathbf{X}(u), \mathbf{X}(v)\rangle\right] = \alpha(u,v), \quad d(u,v) = \frac{1}{p}\mathbb{E}\left[\|\mathbf{X}(u) - \mathbf{X}(v)\|^2\right].$$

$$\max_{i,j\in[n],i\neq j} |m(z_i,z_j) - \hat{m}(i,j)| \leq \max_{i,j\in[n],i\neq j} |\alpha(z_i,z_j) - \hat{\alpha}(i,j)|.$$

$$\max_{i,j\in[n],i\neq j} |\alpha(Z_i,Z_j) - \hat{\alpha}_{\text{data}}(i,j)| \in O_{\mathbb{P}}\left(\frac{n^{2/q}}{\sqrt{p}}\right).$$

$$\max_{i,j\in[n],i\neq j} |\alpha(Z_i,Z_j) - \hat{\alpha}_{\text{pca}}(i,j)| \in O_{\mathbb{P}}\left(\sqrt{\frac{nr}{p}} + \sqrt{\frac{r}{n}}\right).$$

$$\max_{i,j\in[n],i\neq j} |m(Z_i,Z_j) - \hat{m}(i,j)|$$

$$\hat{\alpha}(u,v) = \frac{1}{|u||v|}\sum_{i\in u, j\in v} \hat{\alpha}(i,j), \quad \text{for all distinct pairs } u,v \in P_m.$$

$$m(z_i,z_j) - \hat{m}(i,j) \geq -M.$$

$$\hat{\alpha}(k,l) \geq H - M, \quad \forall k,l \in S_v.$$

$$m(z_i,z_j) - \hat{m}(i,j) \leq M.$$

$$\max_{k\neq l\in[n]} \hat{\alpha}(k,l) \geq \alpha(z_i,z_j) - M,$$

$$\hat{\alpha}(k,l) \leq H - b + M, \quad \forall\, k \in S_v, l \in \bar{S}_v.$$

$$\max_{i\in[n]} \left\|p^{-1/2}\mathbf{Q}\zeta_i - \phi(Z_i)\right\| \in O_{\mathbb{P}}\left(\sqrt{\frac{nr}{p}} + \sqrt{\frac{r}{n}}\right).$$

$$\mathbb{P}\left(\left|p^{-1}\left\langle \mathbf{Y}_i, \mathbf{Y}_j\right\rangle - \alpha(Z_i,Z_j)\right| \geq \delta \middle| Z_1,\ldots,Z_n\right) \leq \frac{1}{\delta^q} C(q,\varphi)\frac{1}{p^{q/2}} M(q,\mathbf{X},\mathbf{E},\mathbf{S})$$

$$\mathbf{Y}_i = \gamma_i \mathbf{X}(i),$$

$$\left|p^{-1/2}\|\mathbf{X}(i)\| - 1\right| \leq \left|p^{-1}\|\mathbf{X}(i)\|^2 - 1\right| = \left|p^{-1}\sum_{j=1}^{p}\Delta_j(i)\right|,$$

$$\begin{aligned}\mathbb{E}[X_j(u)X_j(v)] &= \mathbb{E}\left[\mathbb{E}[X_j(u)X_j(v)|X_j(w)]\right] \\ &= \mathbb{E}\left[\mathbb{E}[X_j(u)|X_j(w)]\mathbb{E}[X_j(v)|X_j(w)]\right] \\ &= \mathbb{E}[X_j(w)^2].\end{aligned}$$

$$m(u,v) = h(w) = \alpha(w,w) = \frac{1}{p}\sum_{j=1}^{p}\mathbb{E}[X_j(w)^2] = \frac{1}{p}\sum_{j=1}^{p}\mathbb{E}[X_j(u)X_j(v)] = \alpha(u,v),$$

$$\begin{aligned}\hat{h}(w) &= \hat{\alpha}(A',B') \\ &= \frac{1}{|A'||B'|}\sum_{k\in A',l\in B'}\hat{\alpha}(k,l) < \hat{H} = \hat{h}(u),\end{aligned}$$

$$\text{return}_i = \frac{p_i^{cl} - p_i^{op}}{p_i^{op}},$$

$$\max_{i,j\in[n],i\neq j}|\alpha(z_i,z_j) - \hat{\alpha}(i,j)| < b/2,$$

$$\hat{\alpha}(A,B) = \frac{1}{|A||B|}\sum_{k\in A,l\in B}\hat{\alpha}(k,l) \geq H - M.$$

$$p^{-1}\langle \mathbf{Y}_i, \mathbf{Y}_j\rangle - \alpha(Z_i,Z_j) = \sum_{k=1}^{4}\Delta_k$$

$$p(x_j) = \prod_{v\in\mathcal{V}} p\left(x_j(v)|x_j(\text{Pa}_v)\right),$$

$$\mathbb{E}\left[X_j(v)|X_j(\text{Pa}_v)\right] = X_j(\text{Pa}_v),$$

$$\hat{\alpha}(w,w') = \frac{1}{|w||w'|}\sum_{i\in w,j\in w'}\hat{\alpha}(i,j).$$

$$\mathbb{E}[\|\mathbf{S}(Z_i)\mathbf{E}_i\|^2|Z_1,\ldots,Z_n] \leq \|\mathbf{S}(Z_i)\|_{\text{op}}^2\mathbb{E}[\|\mathbf{E}_i\|^2] \leq \max_{v\in\mathcal{Z}}\|\mathbf{S}(v)\|_{\text{op}}^2\, p\,\mathbb{E}[|\mathbf{E}_{11}|^2],$$

$$\|\mathbf{Y}_i - \mathbf{Y}_j\|^2 = \|\mathbf{Y}_i\|^2 + \|\mathbf{Y}_j\|^2 - 2\langle\mathbf{Y}_i,\mathbf{Y}_j\rangle,$$

$$\frac{1}{p}\mathbb{E}[\|\mathbf{Y}_i\|^2|\gamma_i] = \gamma_i^2\frac{1}{p}\mathbb{E}[\|\mathbf{X}(i)\|^2] = \gamma_i^2 h(i) = \gamma_i^2,$$

$$\max_{i,j\in[n],i\neq j}|\alpha(i,j) - \hat{\alpha}_{\cos}(i,j)| \in O_{\mathbb{P}}\left(\frac{n^{2/q}}{\sqrt{p}}\right).$$

$$\max_{i\in[n]}\left|p^{-1}\|\mathbf{X}(i)\|^2 - 1\right| \in O_{\mathbb{P}}(n^{1/q}/\sqrt{p}).$$